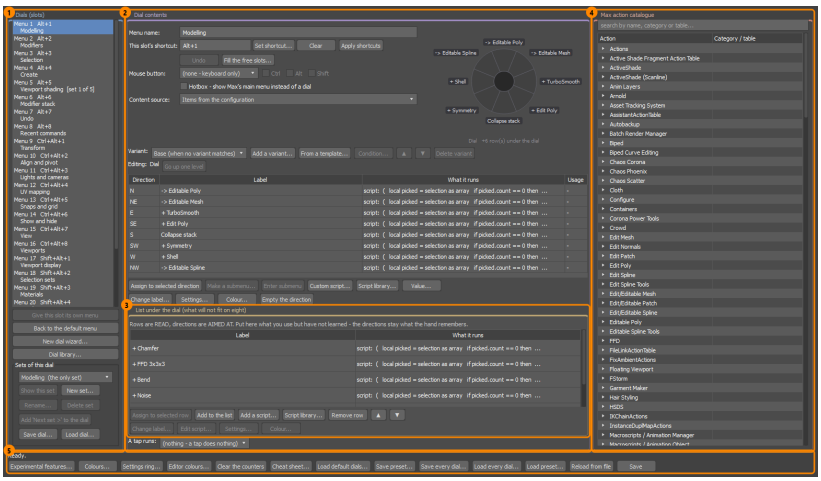
# MarkingForge

Editor Guide - your first dial, and every part of the editor

3DS MAX 2027 · VERSION 1.5.59 · ABOUT 50 MINUTES



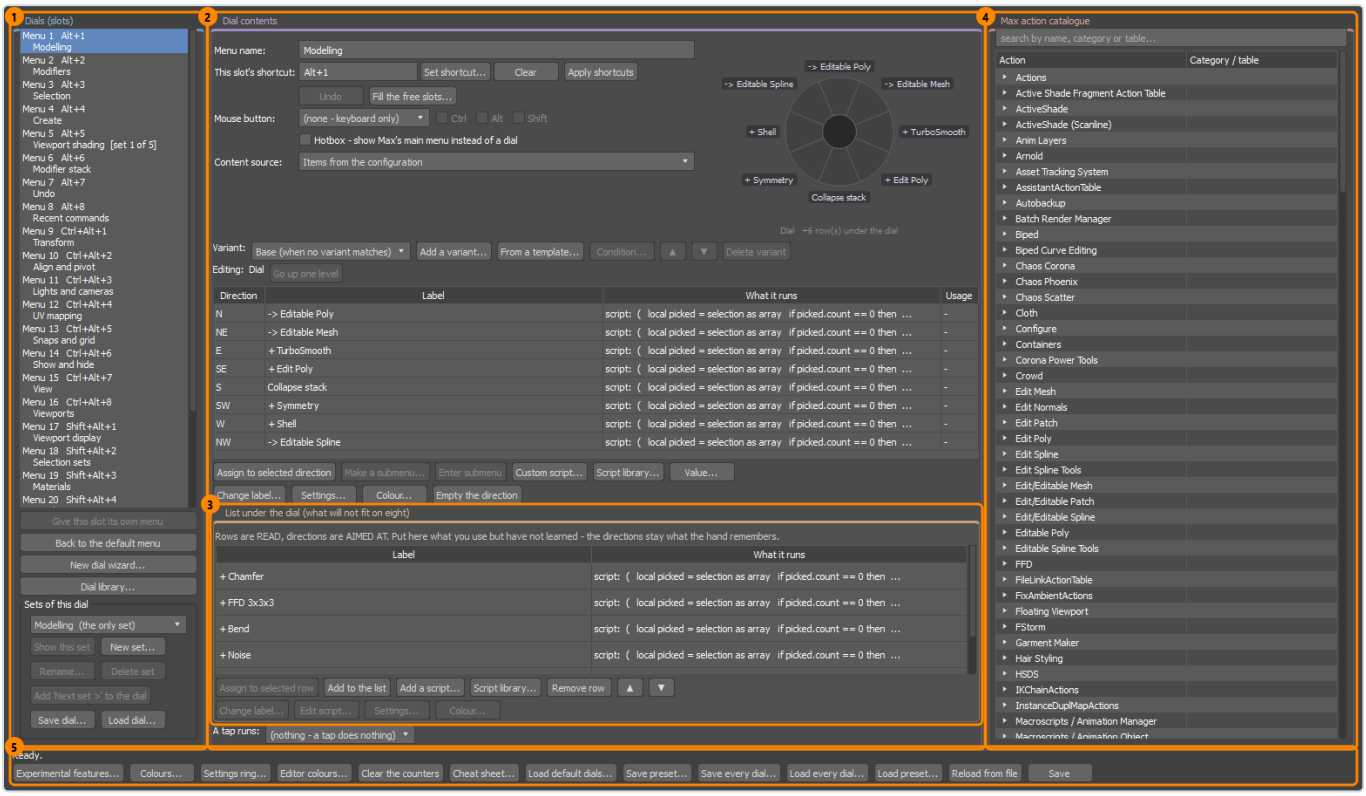
The menu editor decides what every dial holds: the commands on its eight directions, the rows under it, its key, what it shows in which situation, its pages and how it looks. This guide builds a dial from start to finish, then walks through the rest of the window piece by piece. The **QuickStart** (Long or Short) teaches the gestures; the **Reference** lists every option.

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# 1. The editor at a glance

Open it with **MarkingForge > Menu editor...** The title bar names the editor's version and the plugin 3ds Max has loaded - and says *OUT OF DATE* when that plugin is older than the editor needs (restart 3ds Max after an update). Four panes on a neutral grey, each marked by its own colour along its top edge and in its title, and a bar of commands along the bottom:




The editor: Dials (1), Dial contents (2), the list under the dial (3), the action catalogue (4) and the bottom bar (5). The colour bar along each pane's top edge tells the panes apart.

| | Pane | What you do there |
|---|---|---|
| | **Dials (slots)** - left | Pick which of the 24 dials you are editing. Each entry shows the slot, its key and the dial's name - marked *[inherited]* when the slot shows the default menu, *[set 2 of 3]* when the dial has sets. **Right-click** a dial to save or load it (chapter 13). Below the list: **Give this slot its own menu**, **Back to the default menu**, the **New dial wizard**, the **Dial library** (chapter 14) and **Sets of this dial** (chapter 12). |
| | **Dial contents** - middle | A tab for each set when the dial has several (chapter 12). Name, key, mouse button, hotbox, content source and, beside them, a **drawing of the dial**; its variants, where you are (*Editing: Dial* > *variant 2* > *submenu S*), the table of eight directions with the buttons that change them, and what a tap runs. |
| | **List under the dial** - inside the middle | Up to 12 extra rows drawn under the dial, with their own buttons. |
| | **Max action catalogue** - right | Every command 3ds Max and its plugins offer - thousands - searchable by name, category or table. Drag from here onto a direction or a row. |
| | **Bottom bar** | A status line saying what the last thing you did changed, and the commands for the whole file: Save, Reload from file, the dials' colours and the editor's own, experimental features, presets, every dial to files and back, the cheat sheet (chapter 11). |

> **Nothing reaches the dials until you press Save.** Experiment freely; **Reload from file** throws every unsaved change away. Saving reloads the dials in about 150 ms - no need to restart 3ds Max. Closing the editor with unsaved changes asks whether to save them, and keys you set but did not apply are offered for applying.

### A menu of its own, or the default menu

A slot either has a menu of its own or shows the **default menu** - one menu shared by every slot without its own; the list marks such a slot *[inherited]*. After a fresh installation every slot has its own built-in dial. Editing an inherited slot would change the default menu - and every slot that shows it - so the editor asks first.

| Button | What it does |
|---|---|
| **Give this slot its own menu** | Copies the default menu into the slot, to edit it alone. Enabled only on an inherited slot. |
| **Back to the default menu** | DELETES the slot's own menu - and its sets - and the slot shows the default menu again. Save the dial first (**Save dial...**) if you may want it back. |

A content source and a mouse button belong to one dial, so on an inherited slot the editor offers to give it its own menu first; in **Colours...** the *Only* choice stays greyed out until the slot has one.

# 2. Your first dial, start to finish

Every one of the 24 slots comes with a built-in dial, so "your own dial" means taking over a slot. This chapter takes **Menu 20 (Render)** - pick any slot you use least - and turns it into *My tools*: a few commands from the catalogue, a ready-made script, a row under the dial and a key of your choice.

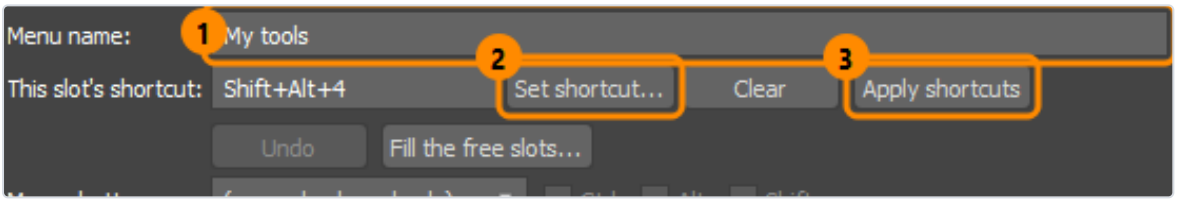
> **In a hurry? New dial wizard...** under the slot list asks three questions - which dial, which eight commands, done - and keeps what the dial showed as another set (chapter 14). The **Dial library...** beside it puts a whole ready-made dial on the slot. This chapter shows the whole road, which both shorten.

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## 1 Pick the slot and name it

1 min · You will learn: where a dial's name shows

1. In **Dials (slots)** click **Menu 20**.
2. In **Menu name** type *My tools*. The name appears in the dial's footer, in the cheat sheet and in 3ds Max's Hotkey Editor next to the slot.




Menu name (1), the slot's shortcut and Set shortcut... (2), Apply shortcuts (3).

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## 2 Give it a key

1 min · You will learn: to open the dial with any keys you like

1. Press **Set shortcut...**; a small window opens. Hold Ctrl, Alt and/or Shift and press a key - for example **Shift+Q**. While the window is open neither 3ds Max nor the dials react to the keys, so even a key that already opens a dial can be recorded. Backspace removes the shortcut, Esc or Cancel leaves it as it was, **Try again** starts over.
2. The window says what already uses the keys: another dial (**Set** moves them here, and that dial is left without a key until you give it one) or a command of your own in 3ds Max (both would react). 3ds Max's built-in defaults are not checked. Press **Set**; **Clear** beside the field leaves the dial without a key.
3. Press **Apply shortcuts**: only then does 3ds Max learn the new key - the shortcuts of all 24 slots are written at once, with a backup, and shortcuts of other commands are kept. **Undo** next to it restores the keys from before.

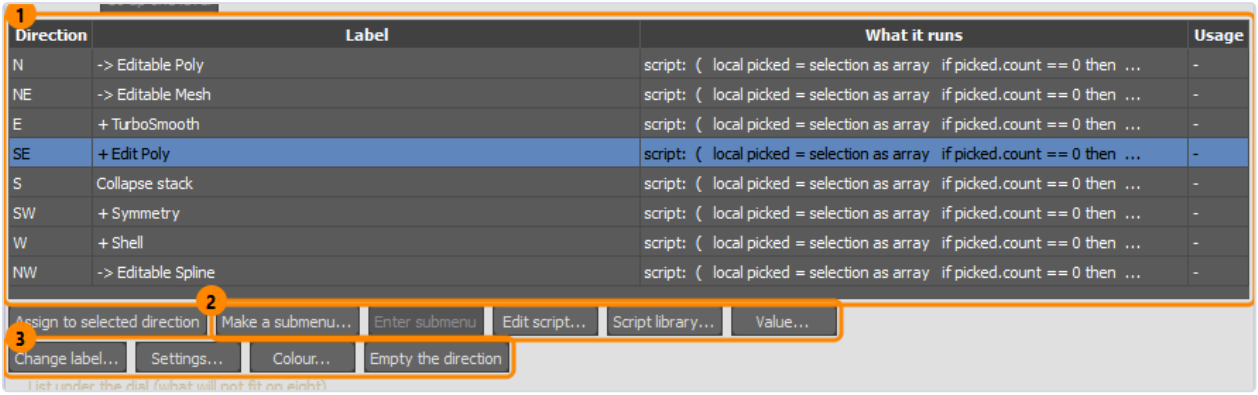
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## 3 Put commands on directions

3 min · You will learn: the one skill everything else builds on

1. Select a direction in the table - say **N** - or click its wedge in the drawing.
2. In the catalogue on the right type part of a command's name - *extrude*. The second column says which table a command comes from: the same name can sit in several (*Select All* is in seven), so pick the one for where you work. Hover over a command for its description and id.
3. Drag the command onto the direction, or double-click it, or press **Assign to selected direction** under the directions table.

4. Do the same for a few more directions. Think in pairs: opposite directions for opposite commands (hide north, unhide south) are the easiest to remember.



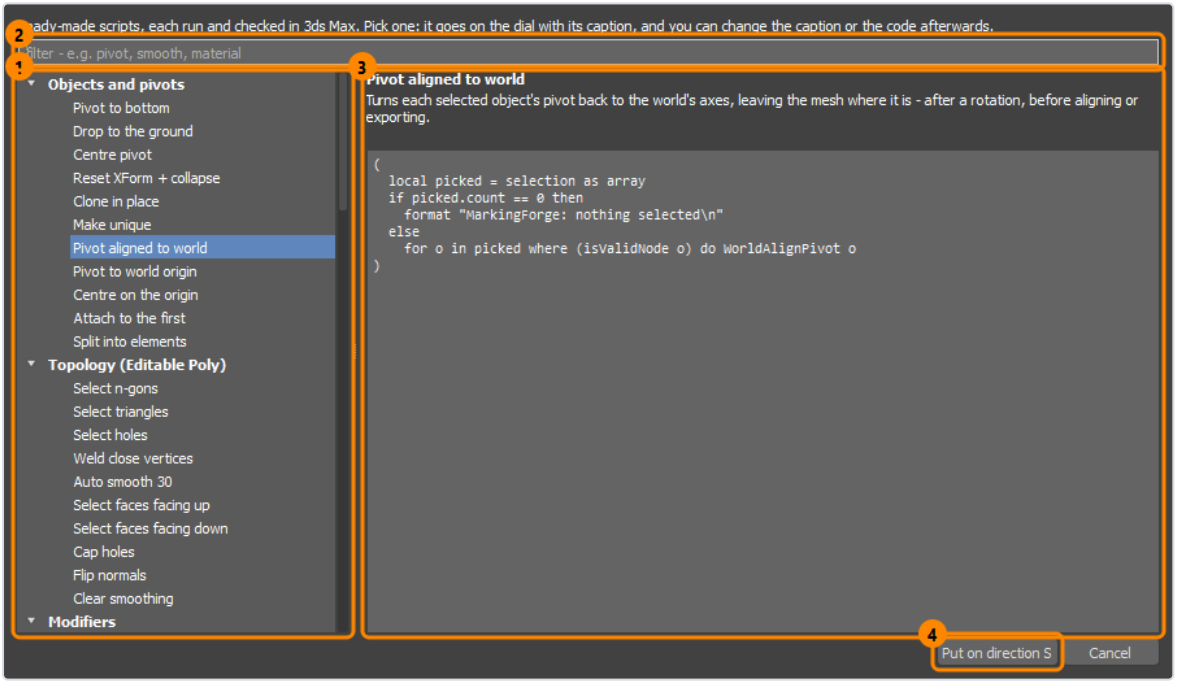

The directions (1) with their buttons: what a direction IS (2), how it LOOKS (3).

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## 4 A ready-made script

1 min · You will learn: commands the catalogue does not have

1. Select an empty direction and press **Script library...**
2. Pick **Pivot to bottom** (or any other) and press **Put on direction N** - the button names the direction you selected.




The script library: the scripts by group (1), the filter (2), what the chosen script does and its code (3), Put on direction (4).

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## 5 Short captions

1 min · You will learn: to keep the dial readable

1. Select a direction with a long name and press **Change label...**
2. Type two words at most - *Extrude* rather than *Extrude Polygons (Editable Poly)*. Leave it empty to take 3ds Max's name again.

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## More than eight: a row under the dial

1 min · You will learn: to keep rarely used commands on the same key

1. Find a command in the catalogue and press **Add to the list** in the list's bar.
2. Rows are read and clicked, directions are aimed at: put here what you use but have not learned.

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## 7 Save and try it

1 min · You will learn: to see the dial

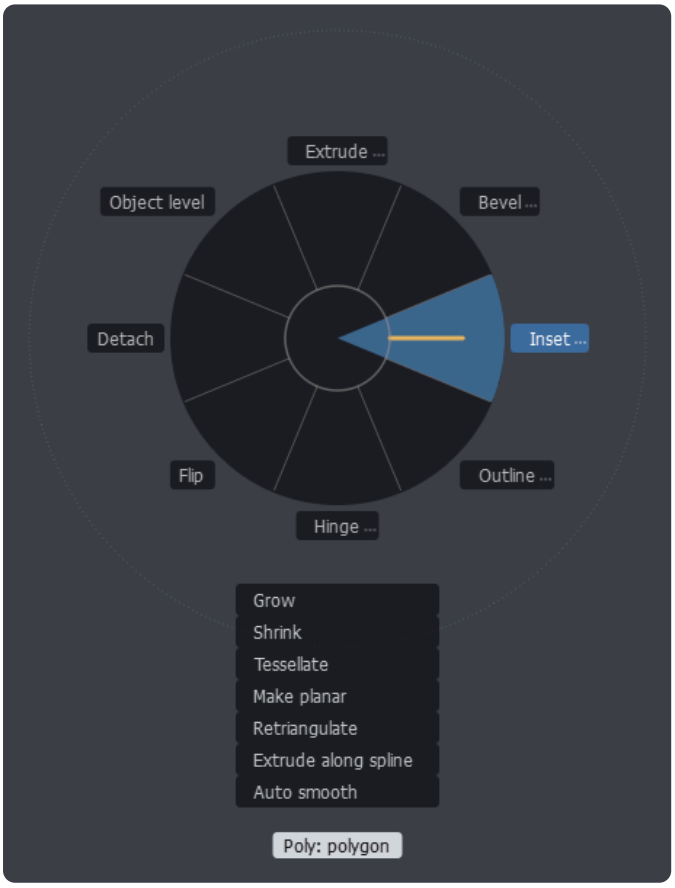
1. Press **Save**.
2. Back in 3ds Max hold your key: *My tools* opens with your commands. Flick towards one to run it without waiting for the dial.

> **RESULT**
>
> A dial of your own, on your key. Everything below makes it smarter: different contents for different objects, the caddy past the rim, submenus, live contents, pages under the mouse wheel, colours.

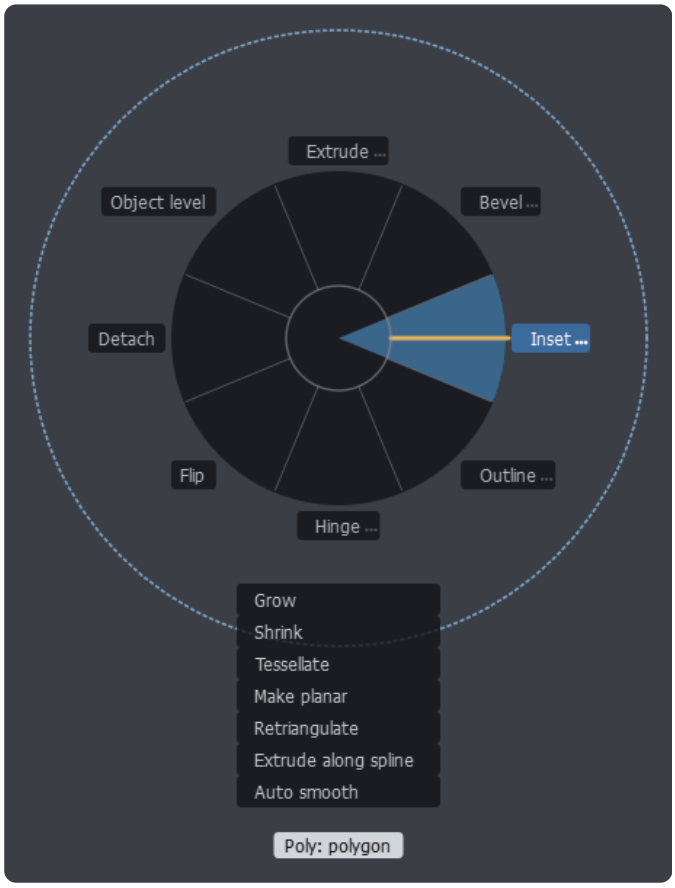
# 3. Rings and circles on the dial

Three round shapes on a dial carry meaning. Each is drawn only when it means something:

| You see | It means | Set in the editor |
|---|---|---|
| **The circle in the centre (36 px in radius)** | Release here to cancel - always, whatever else is going on. It turns to the cancel colour when a release would do nothing. It is large on purpose, so the hand finds it without aiming. | - |
| **A dotted circle outside the dial (the settings rim)** | Commands with a caddy (Chamfer, Extrude, Inset...): an ordinary release does one thing, releasing past this rim the opposite - with or without the caddy. It brightens once you are past it. It lies 20 px beyond the farthest caption (you can move it), so a short movement past the captions is enough. A flick longer than the distance to it counts the same. | **Settings...** under the directions; the global tick box in **Experimental features**; its distance in **Settings ring...** |
| **A dashed circle round the centre** | Multiple picks: go past the rim and come back inside it and the command waits in a queue (a numbered row above the dial). Releasing in this ring (from the centre circle out to 92 px) - green - runs the queue as one undo; the centre circle still cancels. | **Experimental features > Multiple picks in one gesture** (off by default) |




Inside the rim - an ordinary release.



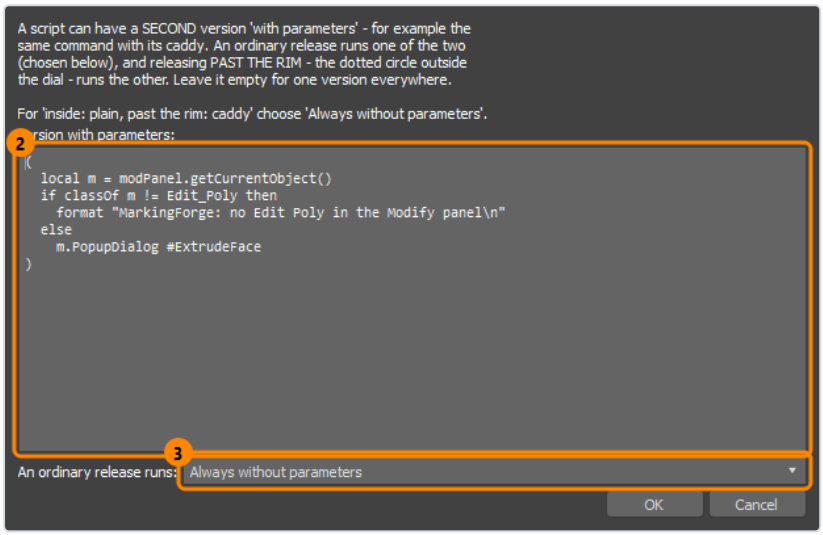

Past the rim - the opposite choice.

### The settings rim, step by step

1. Select a direction with a command that has a caddy - *Chamfer*, say - and press **Settings...**

2. Choose what an ordinary release does: *As the global setting*, *Always with parameters* (the caddy opens) or *Always without parameters* (no caddy - the command starts as its Modify-panel button does; for Chamfer, Inset or Extrude you then drag in the viewport).
3. Past the rim always does the opposite. Save, then try both in a viewport.

The global setting is the tick box **An ordinary release shows the action's SETTINGS** in **Experimental features** - off by default since 1.5.17: an ordinary release runs the command at once, past the rim opens its caddy. On the dial a command with a caddy carries an ellipsis at its right edge, which brightens when the gesture would open the caddy. For a script, **Settings...** takes a second version of the code instead (the library's *Extrude (Edit Poly)* is one); the table marks such a script *[+ with parameters]*:

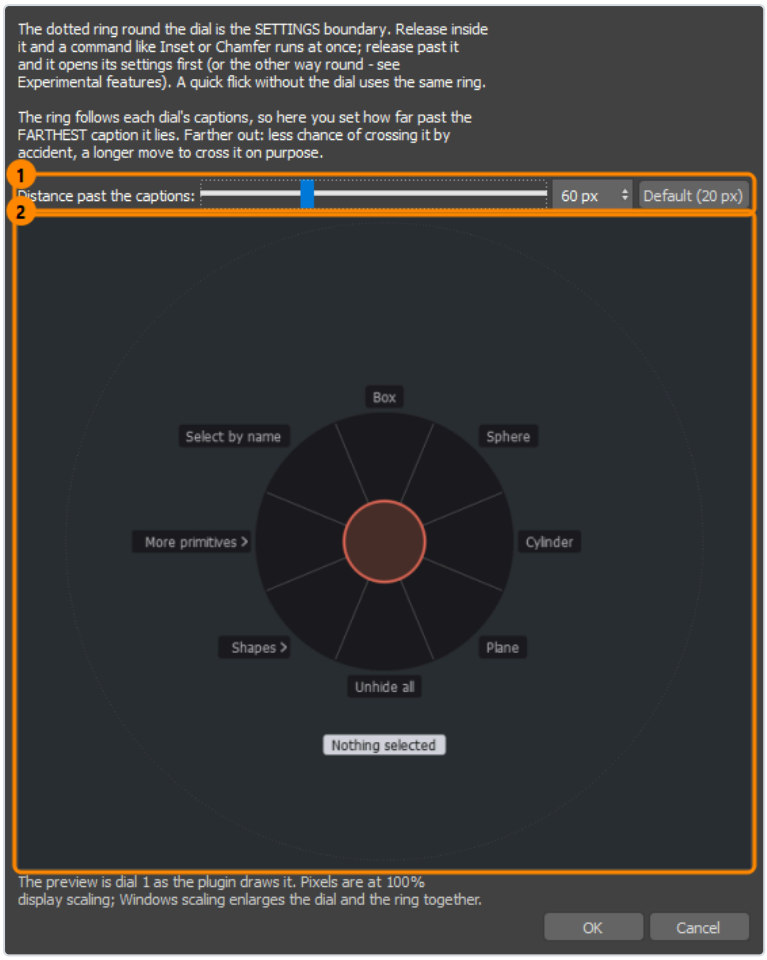


Settings... on a script: its version with parameters.

### The ring too close or too far? Move it

The ring follows the captions of each dial, so you do not set its radius - you set **how far past the farthest caption** it lies. Press **Settings ring...** in the bottom bar (new in 1.5.10):

1. Drag the slider or type a number: 0 to 200 px, 20 px is the default. The picture below it is the real dial of the slot selected in the editor, drawn by the plugin with your number (a hotbox has no ring, so for a hotbox slot the first dial that is not one is shown, and the window says which).
2. Farther out if you cross it by accident (the hand overshoots); nearer if crossing it on purpose feels like a long trip. **Default (20 px)** puts it back.
3. OK, then **Save**. One number for every dial - and for the flick, which is measured against the same ring.

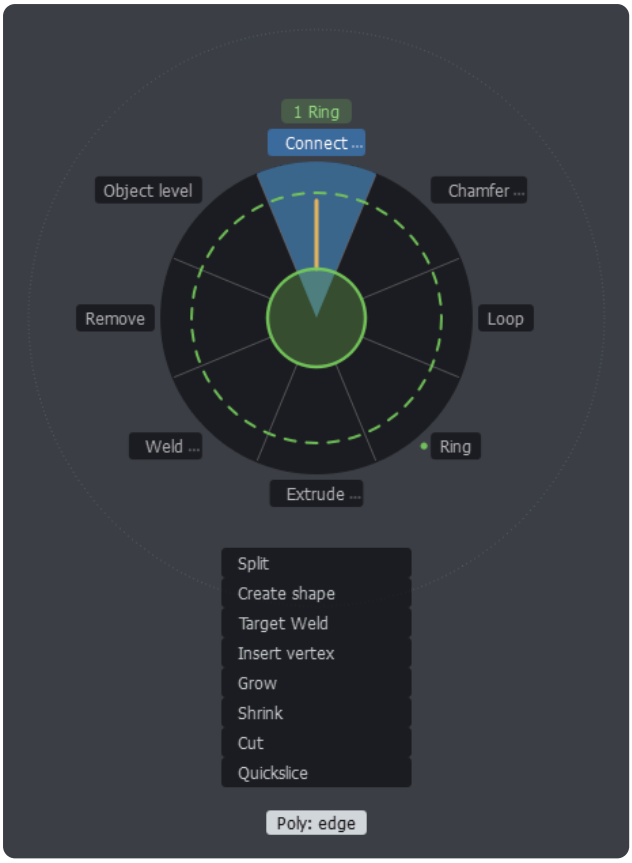



Settings ring...: the distance (1) and the dial with the ring where it will be (2).

### Multiple picks, step by step

1. Open **Experimental features...** and tick **Multiple picks in one gesture**; OK, Save.
2. Hold a key, move past the rim towards a command, come back to the dashed circle without releasing: it is queued. Do it again for another command.
3. Release inside the dashed circle (green): both run, in order, as one undo. Changed your mind? Release on the dot in the middle, or over an empty direction.

Chain commands that finish at once (Ring, Loop, Flip, Connect, Weld, a modifier) as one undo. Inset, Extrude, Bevel and Chamfer wait for a drag in the viewport: the queue stops there until you right-click, then the next command starts (the status line names it). Esc keeps the tool on; another tool, another object or a new gesture drops the rest; each waiting step is its own undo.



A queued command (the row above the dial) and the dashed return circle.

# 4. Directions, button by button

Two rows of buttons under the table. The first changes **what** the selected direction is, the second how it **looks and behaves**. **Enter submenu** is greyed out unless the direction holds one.

| Button | What it does |
|---|---|
| **Make a submenu...** | Turns the direction into another dial with its own eight directions (chapter 7). It reads *(too deep)* on the last level the plugin accepts. |
| **Enter submenu** | Edits the submenu under the direction. Double-clicking the row does the same. |
| **Custom script... / Edit script...** | Your own MAXScript on the direction - or its code, when it already runs a script. Scripts run with full rights. |
| **Script library...** | 102 ready-made scripts, each tested in 3ds Max (chapter 5). |
| **Value...** | A slider: the distance you move sets a number (chapter 8). |
| **Change label...** | A caption of your own. Empty = 3ds Max's name. |
| **Settings...** | With or without the caddy on an ordinary release; for a script, its second version (chapter 3). A value or a submenu has none - the editor says so. |
| **Colour...** | This one direction's colours: its **Caption** and its **Tile**, each with its own tick box *own colour* - a red caption on a destructive command, say. A broken entry stays red and an unavailable one grey whatever you choose. |
| **Empty the direction** | Removes what the direction runs; releasing there cancels. |

The table shows each direction's **Label**, **What it runs** and its **Usage** (chapter 10). The second column says what kind of entry it is:

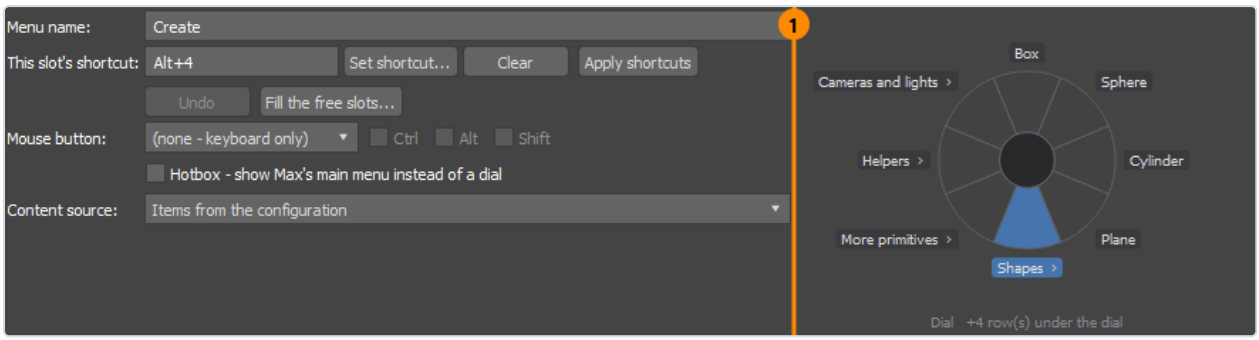
| What it runs reads | The direction holds |
|---|---|
| ***action: Main UI (name from Max)*** | A 3ds Max command - its table, and whether the caption is 3ds Max's own name. |
| ***macro: category / name*** | A macroscript. |
| ***script: [+ with parameters] code...*** | A script; the mark when it has a second version. |
| ***value: 10.0..120.0 every 3.0 px*** | A value slider; its caption ends with **<>**. |
| ***submenu: 5 item(s) (E, N, ...)*** | A submenu; its caption ends with **›**. |
| ***ERROR: ...*** | A command this 3ds Max does not have - a missing plugin, usually. Without a caption of its own the label reads *! CANNOT BE RESOLVED*; on the dial and in the drawing it is red. Replace it. |
| ***(empty - releasing cancels the gesture)*** | Nothing. |

> Putting anything on a direction that holds a submenu asks first: the submenu and everything in it would be replaced, and no button undoes that - only **Reload from file** before you Save.

### The drawing of the dial

Beside the name and the key the editor draws the dial you are editing - north at the top, as it opens. The table says *what* each direction runs; the drawing says *where*, which is how a marking menu is learnt. **Click** a wedge to select that direction in the table; **double-click** a submenu (marked ›) to step inside; **right-click** for the dial's menu - save it, load a file into it, put a library dial on it, or save and load every dial (chapter 13). A caption in red is a command this 3ds Max does not have. Under the drawing: where you are (dial, set, variant, submenu) and how many rows the list under the dial holds; a hotbox or a dial that

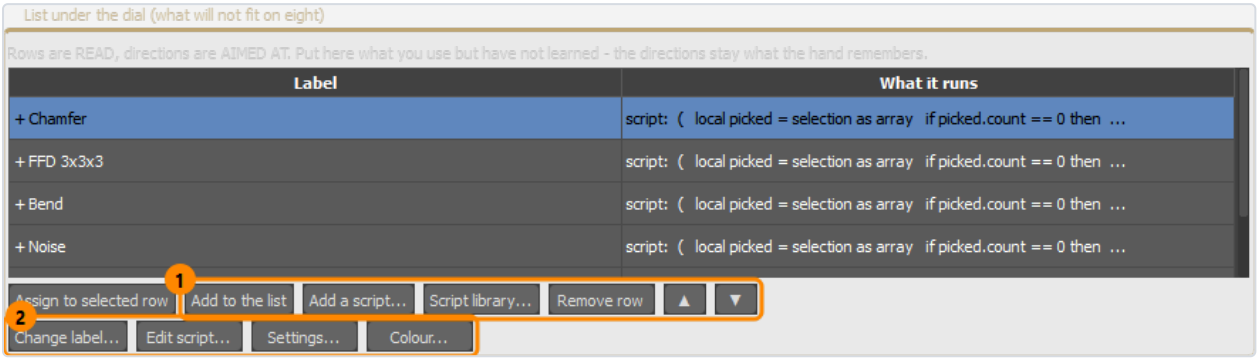
fills itself says so instead of showing directions. In a narrow window the drawing gives way to the form - widen the window or the middle pane to bring it back.




The drawing (1) beside the form: the selected direction in blue, a submenu with ›.

# 5. The list under the dial, and scripts

The list has its own bar, in two rows like the directions' - what is **in** the list above, editing the **selected row** below. Up to 12 rows, drawn under the dial in the order you see:




Filling the list (1) and editing the selected row (2).

| Button | What it does |
|---|---|
| **Add to the list** | Appends the command selected in the catalogue. Dragging onto the list works too - between rows to insert, below the last one to append. |
| **Add a script... / Script library...** | Appends your own script - and asks for its caption at once, since rows are read - or a ready-made one. |
| **Remove row, ▲ ▼** | Removes the selected row, moves it up or down. Order is what you see. |
| **Change label..., Edit script..., Settings..., Colour...** | The same edits a direction has. A row has no tile, so Colour... sets the caption only. A row cannot be a value or a submenu. |

### Ready-made scripts

102 scripts in 11 groups, each run in 3ds Max on a test scene and with nothing selected:

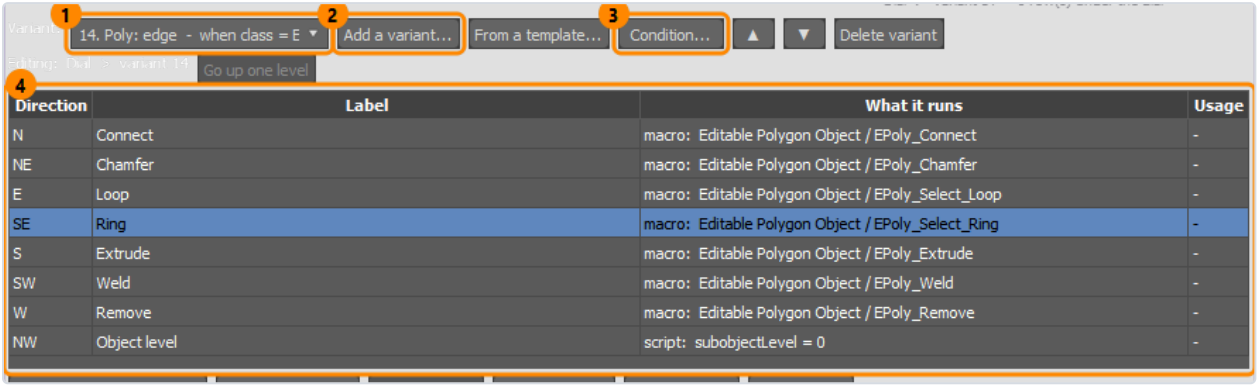
| Group | Scripts |
|---|---|
| **Objects and pivots** | Pivot to bottom, Drop to the ground, Centre pivot, Reset XForm + collapse, Clone in place, Make unique, Pivot aligned to world, Pivot to world origin, Centre on the origin, Attach to the first, Split into elements |
| **Topology (Editable Poly)** | Select n-gons, Select triangles, Select holes, Weld close vertices, Auto smooth 30, Select faces facing up, Select faces facing down, Cap holes, Flip normals, Clear smoothing |
| **Modifiers** | Smoothing on/off, + UVW Map (box), Remove top modifier, Extrude (Edit Poly), + Shell, + Symmetry, + Chamfer, + Edit Poly, + FFD 3x3x3 |
| **Materials and display** | Grey clay material, Remove material, Random wire colour, See-through on/off, Display as box on/off, UV checker material, Material = wire colour, Backface cull on/off, Random colour materials, Merge same-named mats, First's material to all, Wire colour = material |
| **Selection and transforms** | Select same material, Select instances, Copy transform, Paste transform, Reset rotation, Select same type, Select children, Select without material, Random rotation (Z), Random scale (10 %), Spread evenly along X |
| **Shapes and splines** | Renderable on/off, Close all splines, + Extrude 10, Spline from edges, Lighter splines (2), + Lathe, Renderable, 2 thick |
| **Scene and layers** | Selection to a new layer, Freeze selection, Unfreeze all, Zoom to selection, Size to the status bar, Rename in sequence, Layers by object type, Select whole layer, Delete empty layers, Scene report |

| | |
|---|---|
| **Modelling (advanced)** | Select hard edges (30), Select edge loop, Select edge ring, Random 20 % of faces, Extrude faces (5), Bevel faces, Detach faces to object, Make faces planar, Bake TurboSmooth (1), Symmetrize X (baked), Greeble: random extrude, Chamfer edges (0.5), Connect edges (2), Delete half (-X), Bridge borders |
| **UV mapping** | UVW box (100 units), UVW cylindrical (capped), Unwrap + flatten, Copy UVs to channel 2, Checker 2x finer |
| **Layout and placement** | Snap to grid (10), Arrange in a grid, Stack on top, Align bottoms to first, Drop onto surface, Spread first to last, Random position (5), Scatter onto surface |
| **Cameras, lights and render** | Camera from view, Three-point lights, Render size 1920 x 1080, Lights on/off |

Pick one with **Script library...** - under the directions or under the list. The window shows what a script does and its code before anything goes on a dial; the filter at the top finds one by a word (*pivot*, *smooth*, *material*), and a double-click picks it. Once placed, a script is yours: **Change label...** and **Edit script...** change it. The Reference prints the code of the most useful four and the shape a script of your own should have; the QuickStart Long (lesson 14) puts four of them under the Modelling dial step by step.

# 6. Variants - one key, different situations

A dial can have **variants**: the same key shows different contents depending on the object selected (or under the cursor), its sub-object level, the modifier in the Modify panel, the window under the cursor and more. The built-in Modelling dial has a variant for every Editable Poly level.




Variant (1), Add a variant... (2), Condition... (3), the variant's own directions (4).

1. Press **Add a variant...**: it starts as a copy of what you are looking at, and the condition window opens at once.
2. Say when it applies - for example *Object class: Editable Poly* and *Sub-object level: 2 - usually edge* - and press OK. **Condition...** changes it later.
3. With the variant selected in **Variant**, fill its directions like any dial; **Menu name** renames it.
4. Order matters: the first variant whose condition matches wins. Keep specific ones above general ones (▲ ▼). **Base** is what shows when none matches. **Delete variant** removes the one selected; the base stays.

The **Variant** list numbers the variants in the order they are tested and marks the one that would win <= *matches NOW* for what is selected in 3ds Max at this moment - the quickest way to see a new variant hidden behind an earlier one. Variants belong to the dial, not to a submenu: inside a submenu the row reads *Variant (does not apply to submenus)*.

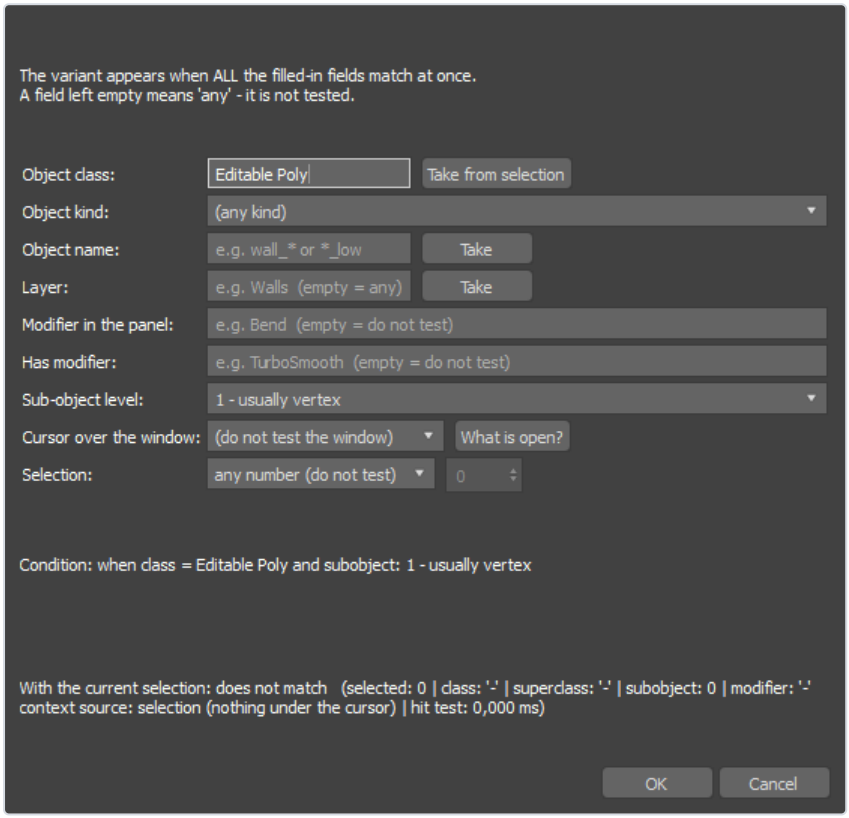
### The condition window

Every field you fill must match at once; a field left empty is not tested.

| Field | The variant appears when |
|---|---|
| **Object class** + **Take from selection** | The selected object is of this class. The button fills class, kind and sub-object level from what is selected now, in the spelling the plugin compares. |
| **Object kind** | The object is any geometry, shape, light, camera or helper - a wider sieve than the class. |
| **Object name** + **Take** | The object's name matches a pattern: *wall_** catches wall_01 and wall_big. Without * the whole name must match. |
| **Layer** + **Take** | The object is on this layer - the whole name, no *. |
| **Modifier in the panel** | The Modify panel is editing a modifier of this name. |
| **Has modifier** | The selected object has this modifier anywhere in its stack. |
| **Sub-object level** | You are on this level: 1 usually vertex, 2 edge, 3 border or spline, 4 polygon, 5 element. |
| **Cursor over the window** + **What is open?** | The cursor is over this window - the 3D view, the command panel, the Slate material editor, the UV editor, Track View, Scene Explorer, Particle View... The button lists the windows open now with the word a rule can use. |

| | |
|---|---|
| **Selection** | Nothing selected, exactly one object, two or more, at least one, or an exact number. |

Under the fields the window says how the plugin reads the condition and whether it *MATCHES* what is selected now - except for *has modifier* and the window, which the editor cannot see and says so. A condition written into menus.json by hand (an object name or a layer) is kept and described. A key 3ds Max does not know (a typo such as `selction`) is kept too, but ignored; a variant whose condition has nothing else is skipped by 3ds Max, and the editor shows it as *SKIPPED*, never as the one that matches.



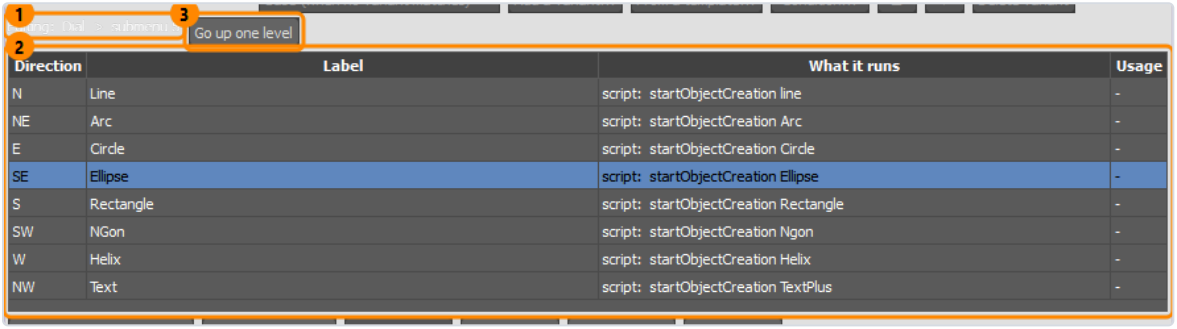

The condition window.

> **From a template...** next to **Add a variant...** adds a ready-made variant instead - one of the built-in dials' context menus with its condition (chapter 14).

# 7. Submenus

A submenu is a whole dial behind one direction: eight more places, reached in two strokes. Use it for a **group** - all the shapes, all the deformers - when the group is too big for the list and too useful to leave out. On the dial a submenu's caption ends with **>**.

1. Select a direction and press **Make a submenu...**; type the caption it will have on the dial. The editor steps inside, and **Menu name** becomes **Submenu caption**.
2. Fill its eight directions. **Go up one level** returns. Double-click a submenu - in the table or in the drawing - to step back in.
3. On the dial, move THROUGH the submenu direction without releasing: the submenu opens IN PLACE - the dial stays where it is and its tiles become the submenu's; move back inside the dial to pick, or into its centre to go up a level. Up to 3 levels, the dial itself counting as the first.




Inside a submenu: the path (1), its directions (2), Go up one level (3).

A submenu has its own directions and list; the key and the mouse button are the dial's, and the hotbox, the content source, the tap and the variants are greyed out inside it.

### Submenus in the built-in dials

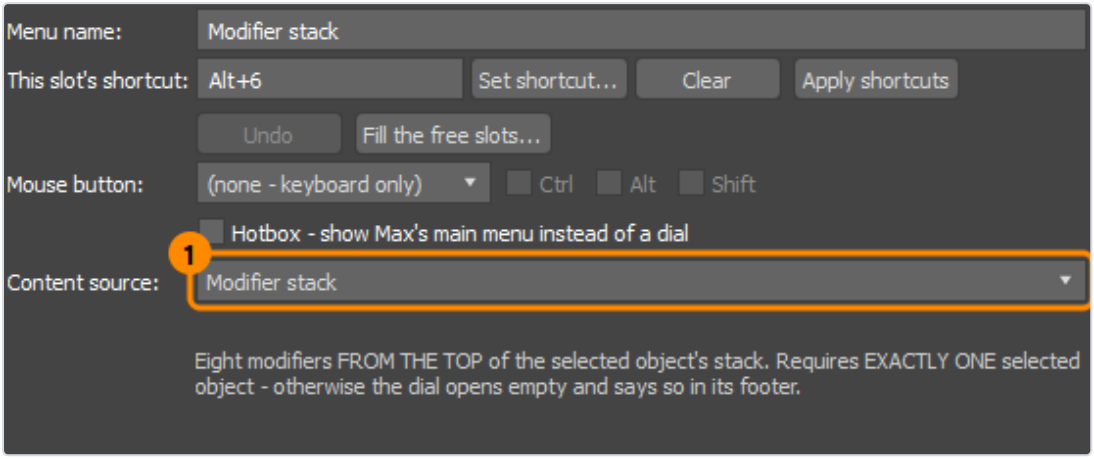
| Dial | Direction | Submenu |
|---|---|---|
| **Alt+4 Create** | S, SW, W, NW | Shapes, More primitives, Helpers, Cameras and lights |
| **Alt+2 Modifiers** | SW, NW | Deform (Bend, Twist, Taper, Noise, FFD, Relax, Stretch, Skew), Geometry (Mirror, Array, Cap Holes, Vertex Weld, Face Extrude, Quadify Mesh, OpenSubdiv, MeshSmooth) |
| **Alt+1 Modelling, nothing selected** | SW, W | Shapes, More primitives - the same submenus as on Create |

The Shapes submenu keeps the directions of the Create dial's shape variant (what Alt+4 shows with a shape selected): Line north, Circle east, Rectangle south - a shape is in the same place either way.

> Dials you already have keep their layout when MarkingForge is updated - nothing is replaced behind your back. **Load default dials...** brings the current built-in dials in: by default into slots without a menu of their own, or, with its tick box, in place of yours - then every dial also gets its default pages and the default dial colours back (your editor colours stay). The **Dial library** (chapter 14) restores one built-in dial alone.

# 8. Content source, sliders, mouse and tap

### Content source - dials that fill themselves



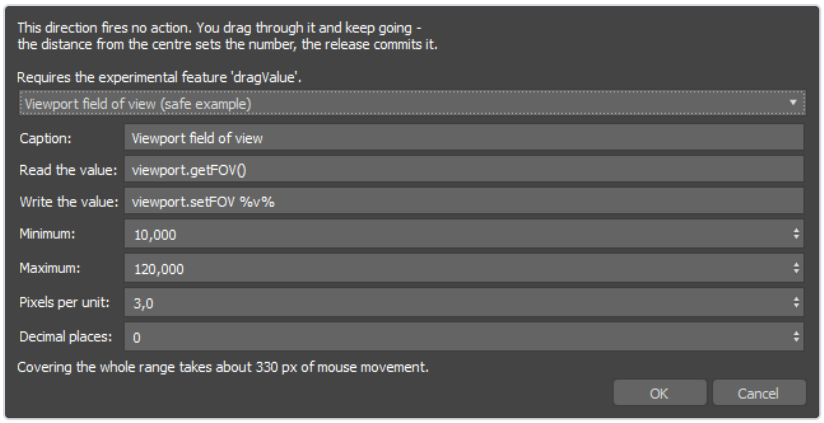

Content source.

| Choice | The dial shows | Releasing on a direction |
|---|---|---|
| **Items from the configuration** | Eight directions exactly as you set them here. An ordinary dial. | runs it |
| **Modifier stack** | Eight modifiers FROM THE TOP of the selected object's stack. Requires EXACTLY ONE selected object - otherwise the dial opens empty and says so in its footer. | makes it current in the Modify panel |
| **Undo stack** | The last eight operations BY NAME. Direction N undoes one, NE two, and so on. Requires the experimental feature 'undoDial'. | undoes back to it |
| **Recent commands** | The eight most recently RUN commands, newest first - from any dial, a tap or the search. For REPEATING, as opposed to the undo stack. The history lives in memory and starts afresh when Max starts. Requires the 'recentDial' toggle. | runs it again |
| **Selection sets** | The scene's named selection sets as eight directions. With more than eight it shows the first eight. Requires the toggle 'selectionSetsDial'. | selects the set |

A source other than *Items from the configuration* fills the directions itself, so the directions, the list and the tap are greyed out and the drawing says *Contents come from the scene*; the sentence under the choice says what the dial will show. Choosing *Undo stack* with its experimental feature switched off offers to switch it on. Lesson 19 of the QuickStart Long sets up each of them step by step. A source can belong to one variant only - the modifier stack only over the command panel, say.

### Value sliders

1. Switch on **Experimental features > Direction that sets a value** - without it the status line reminds you that the direction will do nothing.
2. Select a direction, press **Value...**, pick one of six examples (lesson 15 of the QuickStart Long walks through five of them) - the first, the viewport's field of view, always works - or fill in the caption, how to **read** the value and how to **write** it ( `%v%` stands for the number), the minimum and maximum, pixels per unit and decimal places. The window says how far the mouse must travel to cover the whole range.
3. On the dial move into that direction and keep going: the distance sets the number, the release commits it.



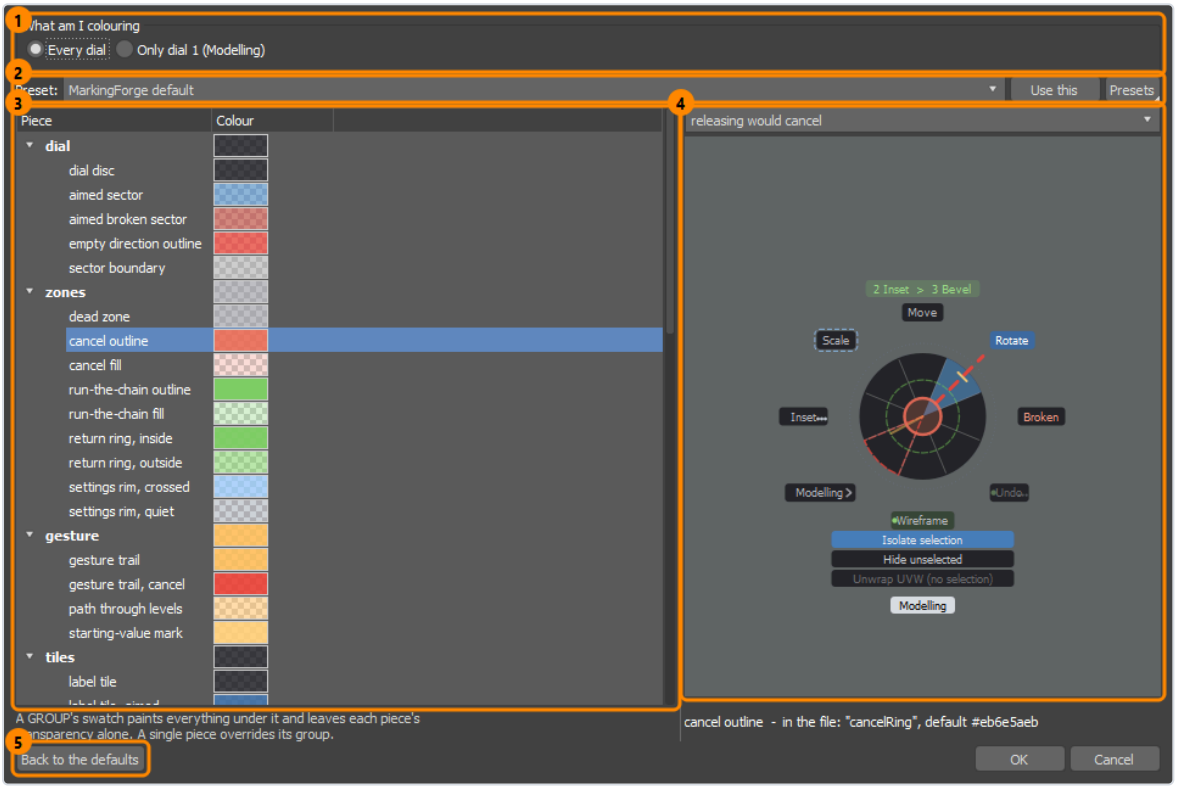
The value window with the viewport field-of-view example.

## Mouse button, tap and hotbox

| Control | What it does |
|---|---|
| **Mouse button** | Opens the dial with a mouse button - left, middle, right or a thumb button - plus Ctrl, Alt and/or Shift, instead of, or as well as, the key. A modifier is required (a bare button belongs to 3ds Max): picking a button ticks Alt if nothing is ticked. A button another dial already uses is reported, since only the first dial would open. |
| **A tap runs** | What a press-and-release WITHOUT movement does: nothing, or one of the directions - through that direction, so changing its command changes the tap too. |
| **Hotbox** | The tick box **Hotbox - show Max's main menu instead of a dial** shows 3ds Max's whole menu bar as tiles instead of a dial (the last page of the built-in Menu 5, Alt+5). The directions and the list are then not used and are greyed out. A hotbox can be one of a dial's sets: the wheel turns to it and away from it like any page. |
| **Fill the free slots...** | Proposes the standard keys (Alt+1-8, Ctrl+Alt+1-8, Shift+Alt+1-8) for every dial that has none, and shows the list - with keys already taken - before writing. |

# 9. Colours

1. **Colours...** in the bottom bar opens the dials' palette. First choose under **What am I colouring**: **Every dial**, or **Only** this dial - its colours go on top of every dial's.
2. Click a group (dial, zones, gesture, tiles, text...) or a single piece: a **group's** swatch paints everything under it and leaves each piece's transparency alone; a single piece overrides its group. The third column marks what you changed.
3. The list above the preview chooses the moment it shows - *a dial standing open*, *aiming at a direction*, *releasing would cancel*, *a chain is waiting*; picking a colour that lives in one moment switches to it. **Back to the defaults** drops every colour of the scope you are in.
4. OK, then Save.




The colour editor: what am I colouring (1), presets (2), the pieces by group (3), the moment and the preview (4), back to the defaults (5).

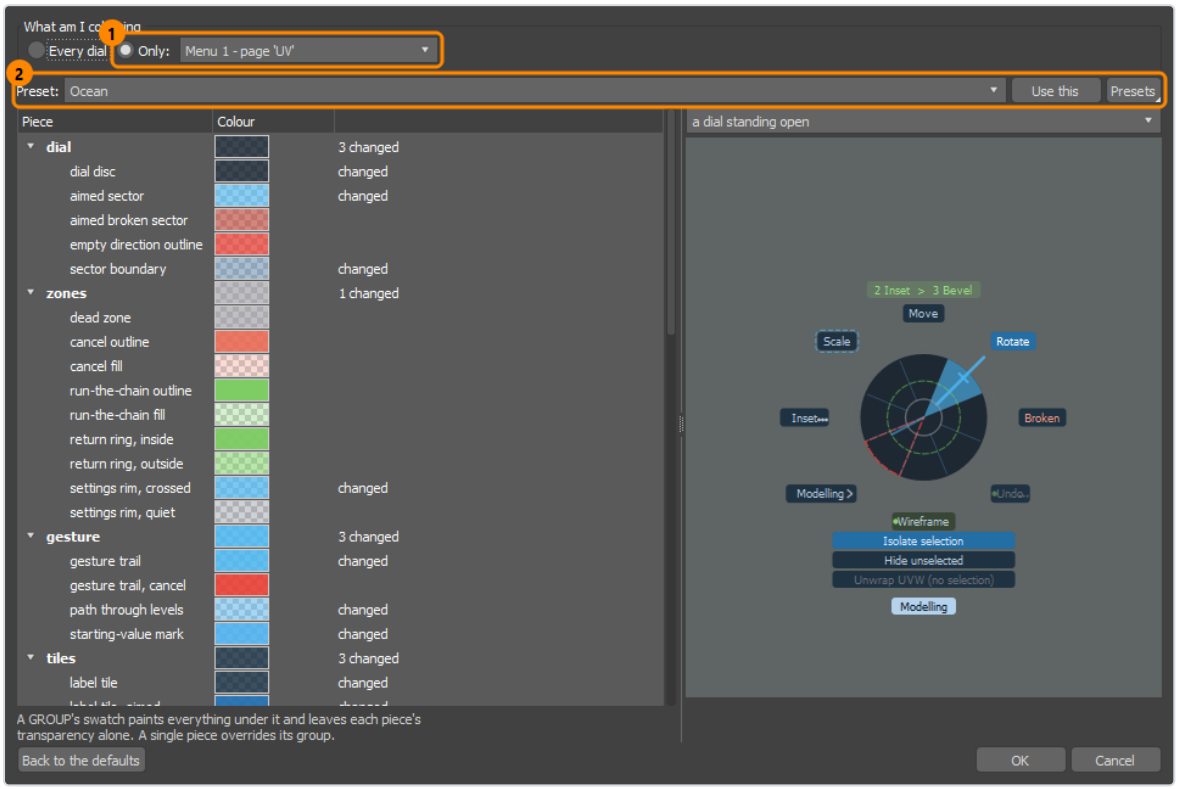
A colour for one command only: select the direction (or row) and press **Colour...** (chapter 4).

## Ready-made looks, your own presets - for every dial, one dial or one page

**Preset** at the top of the colour editor offers 30 ready-made looks - MarkingForge default, Graphite, Slate, Ocean, Violet, Rose, Ember, Mono, Sand - light tiles, High contrast, Indigo, Twilight, Steel, Crimson, Copper, Gold, Ice - light tiles, Lavender - light tiles, Sapphire, Amethyst, Magenta, Coral, Sunset, Amber, Ruby, Storm, Carbon, Neon, Snow - light tiles, Peach - light tiles - and the ones you saved, marked *(yours)*. **Use this** paints what is chosen under **What am I colouring**, replacing the colours it had: **Every dial**, or **Only** one dial. For a dial with sets the list beside **Only** offers **every page** of it and **each page** on its own - the sets the mouse wheel turns; it starts on the page whose tab is open. A dial's colours reach its context variants and its submenus as well, and a page's colours travel with it when it goes on the key. OK writes only what you changed - a page you only looked at keeps its colours.

1. Choose the scope: every dial, one dial, every page of it, or one page.
2. Pick a preset and press **Use this** - or click swatches.

3. **Presets** > **Save these colours as a preset...** keeps the look under a name; **Delete this preset** removes one of yours; **Export this preset to a file...** and **Import a preset from a file...** move one to another computer as a `.mfcolors` file. An imported name you already have gets a number - *Mine (2)*; a dial preset cannot be imported as an editor one, or the other way round.
4. OK, then Save.



One page of Menu 1 (1) in the Ocean preset (2).

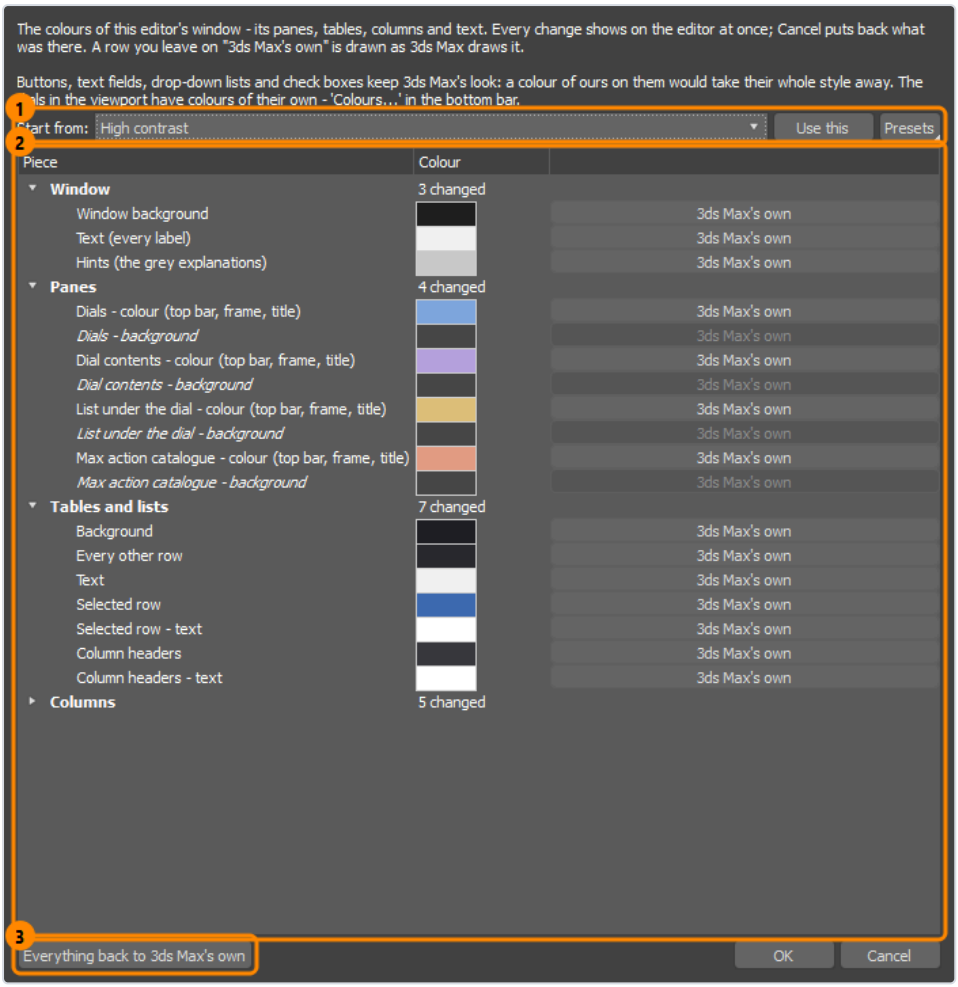



Every ready-made dial look, as the colour editor's preview draws it.

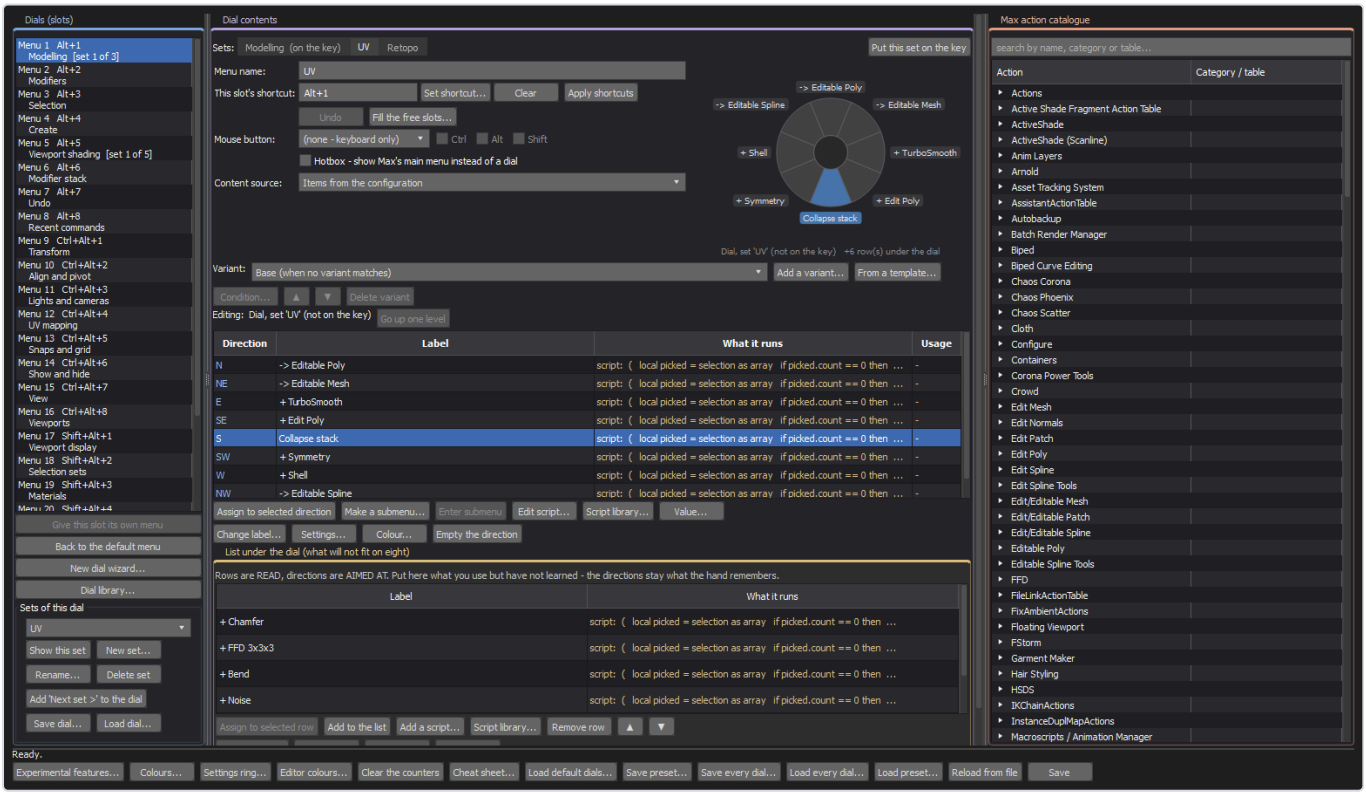
## The editor's own colours

**Editor colours...** in the bottom bar colours this window, not the dials: the window's ground, its text and hints, each pane's colour and ground, the tables and lists - their ground, every other row, text, the selected row and the column headers - and **every column of every table on its own**, text and ground. Nothing is set until you choose it; the editor changes as you pick.

1. Pick a starting point under **Start from** - 21 ready-made looks: High contrast, Calm grey, Graphite, Midnight blue, Warm sepia, Plum, Ocean, Charcoal, Steel blue, Indigo night, Mocha, Rose dust, Nord, Sapphire, Amethyst, Carbon, Storm, Burgundy, Slate grey, Dusk, Espresso - or one of your own, and press **Use this**; *3ds Max's own* starts from nothing. **Presets** saves, deletes, exports and imports your own, as for the dials.
2. Click any swatch in **Window**, **Panes**, **Tables and lists** or **Columns**. A piece in italics is drawn as 3ds Max draws it.
3. **3ds Max's own** beside a row drops that colour; **Everything back to 3ds Max's own** drops them all. **OK** keeps them - no Save needed; **Cancel** puts back what was there.




Editor colours: starting points (1), every piece of the window (2), back to 3ds Max's own (3).

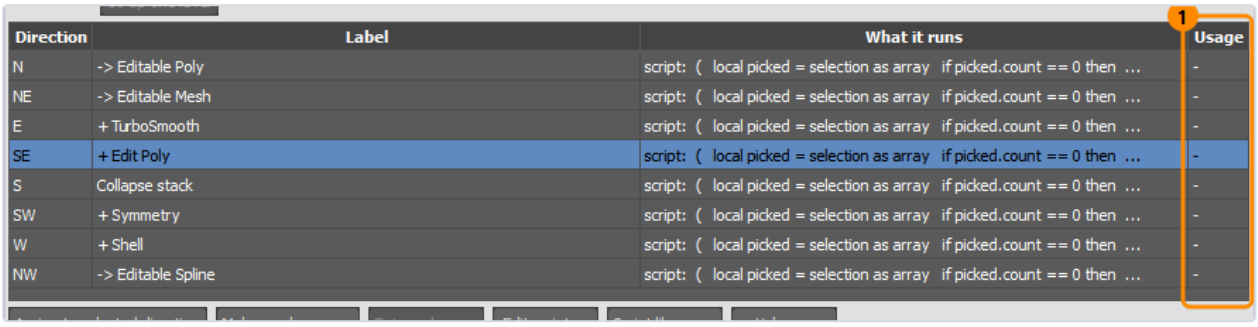



The editor in the High contrast starting point.

> Buttons, text fields, drop-down lists and check boxes keep 3ds Max's look - a colour of ours on them would take their whole style away. The colours live beside menus.json: the editor's own in `editor_colors.json` (delete it to start over), your presets in `dial_color_presets.json` and `editor_color_presets.json`.

# 10. Reading the Usage column

The last column of the directions table counts how you use the dial - quick gestures and held dials alike. *12 x* means the direction ran twelve times; *cancelled 3* that you aimed at it three times and nothing ran - on an empty direction it is drawn in orange, a hint that something is missing there; - that it was never used. The counters are kept per variant and per submenu: choose the variant you worked in to see its numbers. They describe the set on the key, so another set's tab shows none.




The Usage column (1) and the conclusions drawn from it (2).

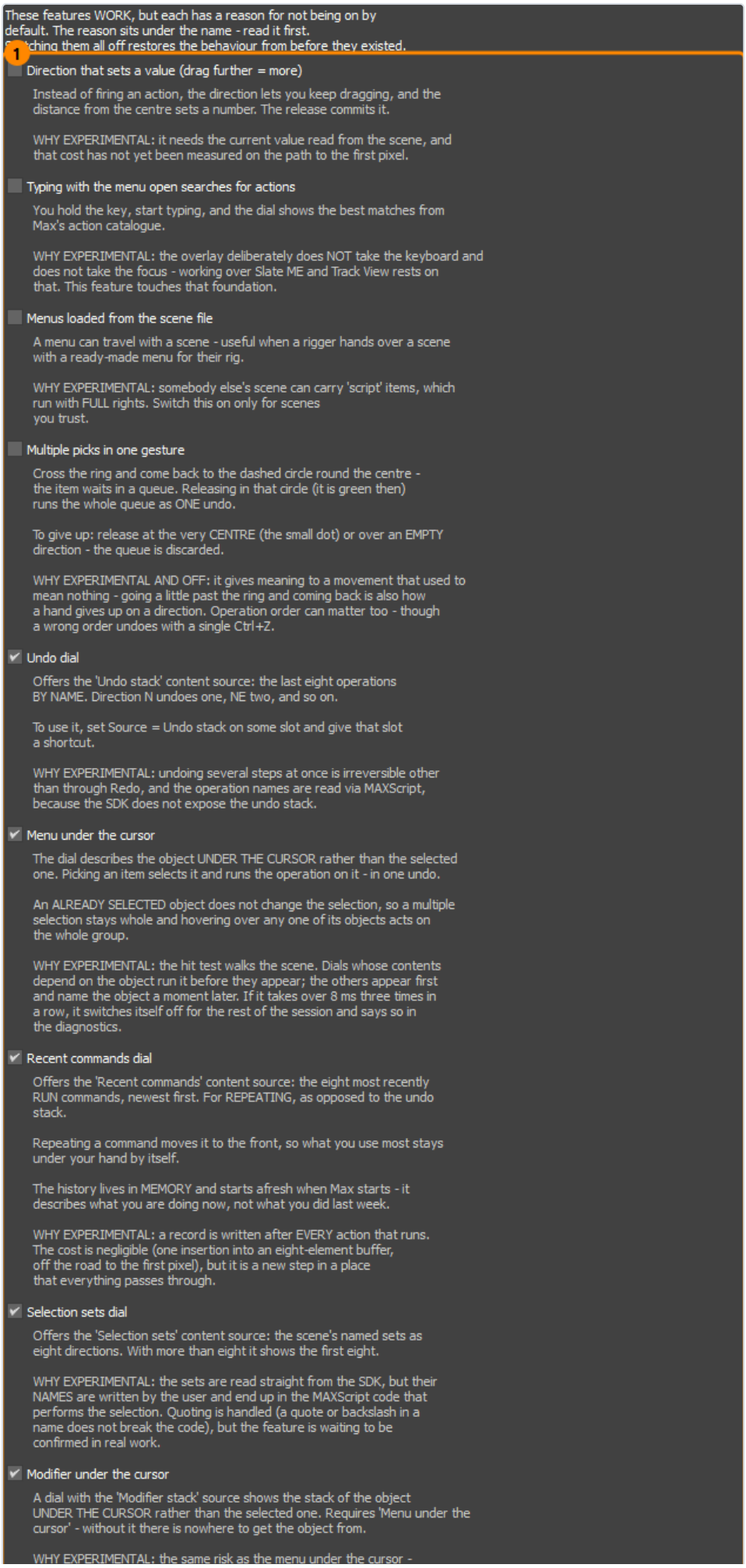
Under the table the editor writes **conclusions**: an empty direction you aim at from the first cancel, and once there are 20 picks or more, a favourite command sitting on a diagonal while a straight direction stands free, a direction never used, more cancels in the centre than picks (a key firing by accident). It never rearranges anything by itself: the hand remembers directions, and a dial that reorders itself would undo that. The editor reads the counters when it opens; **Reload from file** refreshes them, **Clear the counters** starts afresh.

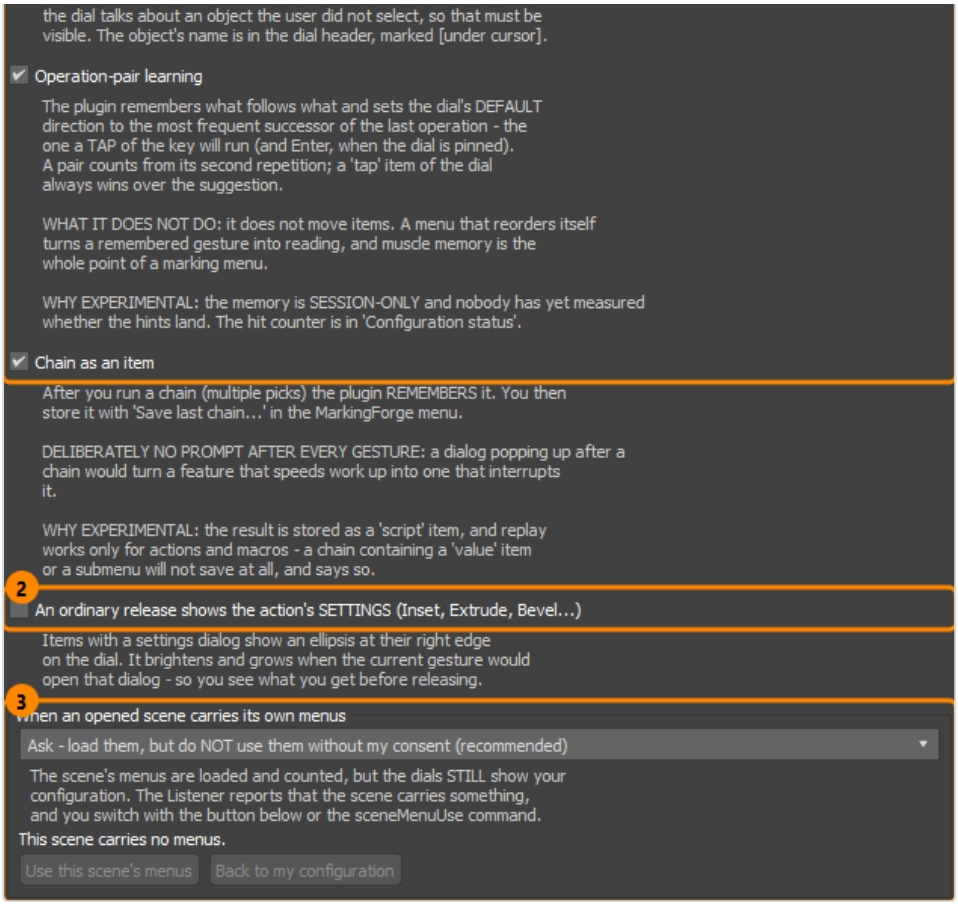
# 11. Experimental features and the bottom bar

**Experimental features...** switches features on and off. Each states in the window why it is experimental - read it before switching one on. Out of the box:

| Feature | Default |
|---|---|
| **Direction that sets a value (drag further = more)** | off |
| **Typing with the menu open searches for actions** | off |
| **Menus loaded from the scene file** | off |
| **Multiple picks in one gesture** | off |
| **Undo dial** | on |
| **Menu under the cursor** | on |
| **Recent commands dial** | on |
| **Selection sets dial** | on |
| **Modifier under the cursor** | on |
| **Operation-pair learning** | on |
| **Chain as an item** | on |

Below the features, two settings that are not experimental: the tick box **An ordinary release shows the action's SETTINGS** (chapter 3; off by default since 1.5.17), and **When an opened scene carries its own menus** - *Ask* (the default: load them, but use them only when you say so), *Always* or *Never* - with **Use this scene's menus** and **Back to my configuration** for the scene open now. OK, then Save.


These features WORK, but each has a reason for not being on by
default. The reason sits under the name - read it first.
Switching them all off restores the behaviour from before they existed.
1
Direction that sets a value (drag further = more)
Instead of firing an action, the direction lets you keep dragging, and the distance from the centre sets a number. The release commits it.
WHY EXPERIMENTAL: it needs the current value read from the scene, and that cost has not yet been measured on the path to the first pixel.
Typing with the menu open searches for actions
You hold the key, start typing, and the dial shows the best matches from Max's action catalogue.
WHY EXPERIMENTAL: the overlay deliberately does NOT take the keyboard and does not take the focus - working over Slate ME and Track View rests on that. This feature touches that foundation.
Menus loaded from the scene file
A menu can travel with a scene - useful when a rigger hands over a scene with a ready-made menu for their rig.
WHY EXPERIMENTAL: somebody else's scene can carry 'script' items, which run with FULL rights. Switch this on only for scenes you trust.
Multiple picks in one gesture
Cross the ring and come back to the dashed circle round the centre - the item waits in a queue. Releasing in that circle (it is green then) runs the whole queue as ONE undo.
To give up: release at the very CENTRE (the small dot) or over an EMPTY direction - the queue is discarded.
WHY EXPERIMENTAL AND OFF: it gives meaning to a movement that used to mean nothing - going a little past the ring and coming back is also how a hand gives up on a direction. Operation order can matter too - though a wrong order undoes with a single Ctrl+Z.
Undo dial
Offers the 'Undo stack' content source: the last eight operations BY NAME. Direction N undoes one, NE two, and so on.
To use it, set Source = Undo stack on some slot and give that slot a shortcut.
WHY EXPERIMENTAL: undoing several steps at once is irreversible other than through Redo, and the operation names are read via MAXScript, because the SDK does not expose the undo stack.
Menu under the cursor
The dial describes the object UNDER THE CURSOR rather than the selected one. Picking an item selects it and runs the operation on it - in one undo.
An ALREADY SELECTED object does not change the selection, so a multiple selection stays whole and hovering over any one of its objects acts on the whole group.
WHY EXPERIMENTAL: the hit test walks the scene. Dials whose contents depend on the object run it before they appear; the others appear first and name the object a moment later. If it takes over 8 ms three times in a row, it switches itself off for the rest of the session and says so in the diagnostics.
Recent commands dial
Offers the 'Recent commands' content source: the eight most recently RUN commands, newest first. For REPEATING, as opposed to the undo stack.
Repeating a command moves it to the front, so what you use most stays under your hand by itself.
The history lives in MEMORY and starts afresh when Max starts - it describes what you are doing now, not what you did last week.
WHY EXPERIMENTAL: a record is written after EVERY action that runs. The cost is negligible (one insertion into an eight-element buffer, off the road to the first pixel), but it is a new step in a place that everything passes through.
Selection sets dial
Offers the 'Selection sets' content source: the scene's named sets as eight directions. With more than eight it shows the first eight.
WHY EXPERIMENTAL: the sets are read straight from the SDK, but their NAMES are written by the user and end up in the MAXScript code that performs the selection. Quoting is handled (a quote or backslash in a name does not break the code), but the feature is waiting to be confirmed in real work.
Modifier under the cursor
A dial with the 'Modifier stack' source shows the stack of the object UNDER THE CURSOR rather than the selected one. Requires 'Menu under the cursor' - without it there is nowhere to get the object from.
WHY EXPERIMENTAL: the same risk as the menu under the cursor -




Experimental features: the features with their reasons (1), the settings rim's global setting (2), menus carried by scenes (3).



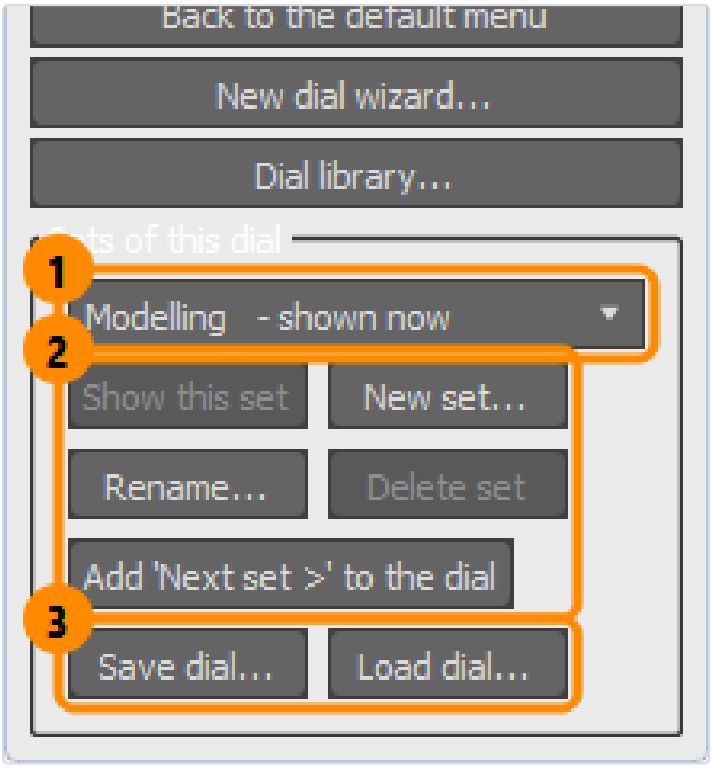

The bottom bar: the status line (1), the buttons (2) - from Assign to selected to Save, as in the table below - and the two for every dial at once (3).

| Button | What it does |
|---|---|
| **Experimental features...** | The window above. |
| **Colours... / Editor colours...** | The dials' palette; this window's own colours (chapter 9). |
| **Settings ring...** | How far the dotted settings ring lies past the captions, with a picture of the real dial (chapter 3). |
| **Clear the counters** | Zeroes the Usage column (chapter 10). The dials are not changed. |
| **Cheat sheet...** | Every SAVED dial with its keys and mouse buttons, sorted by key - a dial with pages with every page, in the order the mouse wheel reaches them - as `MarkingForge_cheatsheet.html` and `.pdf` beside menus.json - with a list of what to check (a dial that would show empty, a dial with no key) and **Open the HTML**. Unsaved changes are not in it: Save first. |
| **Load default dials...** | The built-in layout back - by default only for slots without their own menu; it lists which slots it will fill before it changes any. With *Replace the slots...* ticked, every dial also gets its default pages and the default dial colours back. |
| **Save preset...** | Your layout in one `.json` file, to share or keep: what each dial shows, its pages, the colours of the dials and the default menu - no counters, no paths, not the editor's own colours (chapter 13). |
| **Save every dial... / Load every dial...** | Each dial and each of its sets in a file of its own, and the whole setup back exactly as saved - options included (chapter 13). |

| | |
|---|---|
| **Load preset…** | Shows what the file holds first, then replaces the slots it contains - with their pages. A preset of your whole layout also sets the dial colours as saved; a pack of a few dials leaves them alone. Scripts in it are removed unless you press **Load WITH the scripts**. |
| **Reload from file** | Throws unsaved changes away (it asks first) and reads the dials, the keys, the counters and the catalogue afresh. |
| **Save** | Writes the dials; the plugin reloads them at once. If the file changed on disk meanwhile - a set switched from a dial, say - Save asks: **Overwrite with my edits**, **Save a copy…** or **Cancel** (1.5.55). |

# 12. Sets - several dials on one key

A dial can keep several **sets** of contents under names - *Modelling*, *UV*, *Retopo* - and switch between them in place. The set the dial shows is an ordinary dial; the others wait beside it in the same file. The key, the mouse button and the other 23 dials stay as they are.



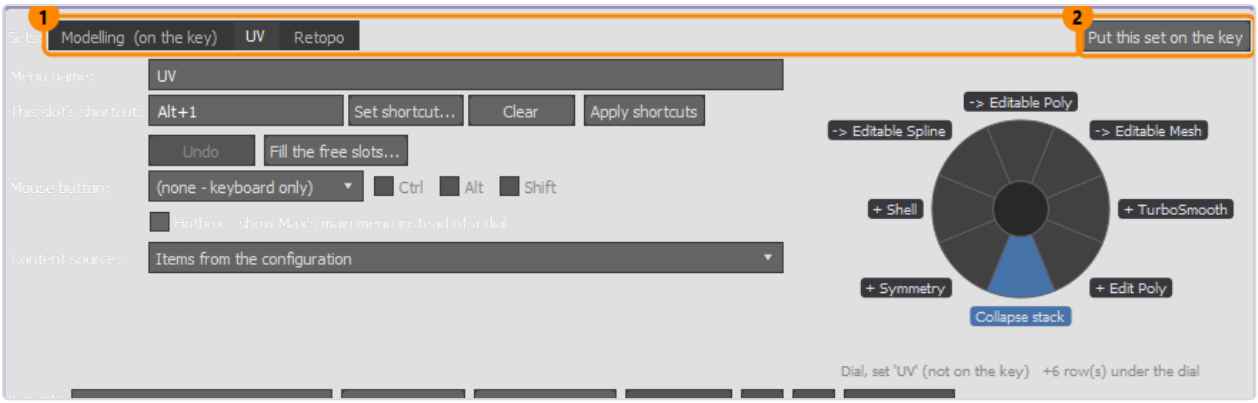

Sets of this dial: the sets (1), switching and managing them (2), one dial in a file (3).

| Button | What it does |
|---|---|
| **New set...** | Asks for a name, then whether the set starts as *A copy of what the dial shows now* or *An empty dial*, and opens its tab. The key keeps opening what it opened; the new set is the dial's next page. |
| **Show this set** | Makes the set chosen in the list the one the dial shows; the one it showed is kept under its name. |
| **Rename... / Delete set** | Renames the chosen set; deletes it. The set the dial shows cannot be deleted - show another one first. With one set left, the dial is simply a dial again. |
| **Add 'Next set >' to the dial** | Puts a *Next set* > row on the list under the dial in every set (and their variants), so you can switch from the dial and come back. |
| **Save dial... / Load dial...** | One dial in a `.mfdial` file: save the set chosen in the list; load a file as a new set or in place of what the dial shows (chapter 13). |

The list names the set the key opens - *shown now* and follows the tab you have open, so Rename, Delete and Save dial act on the set you are looking at. A dial without sets shows itself there as *(the only set)*.

### Looking at another set - tabs and the mouse wheel

Over the dial's contents each set is a **tab**; the one the key opens is marked *(on the key)*. Click another tab to see and edit that set - the drawing, the directions, the list and the variants are all that set's, and the path reads *Dial, set 'UV' (not on the key)* - while the key keeps opening the set it opened. **Put this set on the key** changes that. With one set the tabs are hidden.




The sets as tabs (1): another set open for editing, and the button that puts it on the key (2).

In 3ds Max the sets are the dial's **pages**: hold its key and **turn the mouse wheel** - one page per notch, the next set (wheel towards you) or the previous one, round and round, at the same place. Release over a command to run it. The next press opens the page you turned to last - also after 3ds Max restarts (1.4.8). The footer says *page 2 of 3 - mouse wheel*. The wheel is taken only when it turns a page - not in a submenu, on a pinned dial or while searching; at any other time it zooms and scrolls as always.

### Ways to switch the set on the key

| Where | How |
|---|---|
| **In the editor** | The set's tab and **Put this set on the key** - or pick it in **Sets of this dial** and **Show this set**. Save. |
| **On the dial** | The *Next set* > row (see above). The dial changes in a moment and the status bar says which set it shows - *Menu 1 -> set 'UV' (2 of 3)*. |
| **Under a key** | **Customize > Hotkey Editor**, category **MarkingForge**: **Menu 1 - next set** ... **Menu 24 - next set**. Give the one you use a key. |

> A switch from the dial or a key writes the file at once. If the editor is open with unsaved changes, press **Reload from file** before you go on - Save would find the file changed and ask whether to overwrite the switch with your edits, keep them as a separate copy, or cancel.

# 13. Dials in files - one dial, every dial, a preset

Three ways to take dials out of the editor and bring them back, from one dial to the whole layout. Nothing that is loaded reaches 3ds Max until you press Save.

| To save | To load | What travels |
|---|---|---|
| **Save dial... (Sets of this dial), or right-click the dial** | **Load dial...**, or right-click the dial | One dial in a `.mfdial` file - the set chosen in the list, with its variants, submenus, list and colours, without its mouse button. Loading asks: *A new set (keeps what the dial shows now)* or *Replace what the dial shows*. |
| **Save every dial... (bottom bar, or right-click)** | **Load every dial...** | Every dial: in the folder you choose a new one, *MarkingForge dials <date time>*, with a file per dial and per other set (*01 Modelling.mfdial*, *01 UV.mfdial*...) - so any one can also go back alone with **Load dial...** - and a list of them, *MarkingForge dials.json*, which since 1.5.55 also keeps the whole configuration: the default menu, the mouse buttons and every option. |
| **Save preset...** | **Load preset...** | What each dial shows, its other sets (pages), the colours of the dials and the default menu, in one `.json` file to share. Loading it replaces the pages the slots have now. Your editor colours are not in it. |

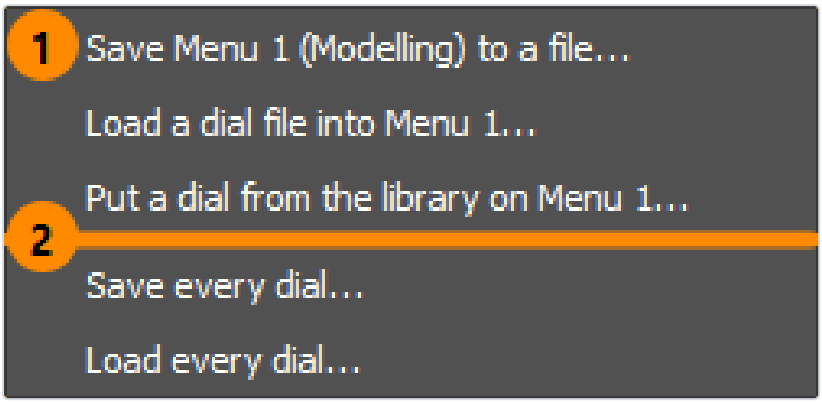
### Every dial back at once

1. Press **Load every dial...** and choose a folder **Save every dial...** made - the dated one, or the folder you saved INTO: its newest *MarkingForge dials* folder is used (1.5.55).
2. If the dials in it run scripts, choose **Remove the scripts** (the default) or **Keep them - I trust these files**.
3. The editor asks once more. A folder saved with 1.5.55 or later puts back EXACTLY what was saved - every dial, its sets in their order (the pages under the mouse wheel), the default menu, the mouse buttons, the colours and every option (settings ring, Settings..., experimental features, scene menus); changes made since are dropped. Keyboard shortcuts live in 3ds Max's own hotkey file and are not part of it. An older folder puts back the dials, their sets and the colours. A damaged file stops the whole load, so you never get half of one layout and half of another.
4. Save.

A folder saved before 1.4.4 has no list; it is read from the file names, the first file of each dial being the set it showed.

### The right-click menu of a dial

Right-click a dial in the list of dials, or its drawing, for the same commands where the dial is - the menu acts on the dial you clicked:



Right-click a dial: this dial to a file, a file or a library dial onto it (1); every dial (2).

| Entry | Does what |
|---|---|
| **Save Menu N (name) to a file…** | **Save dial…** |
| **Load a dial file into Menu N…** | **Load dial…** |
| **Put a dial from the library on Menu N…** | **Dial library…** (chapter 14) |
| **Save every dial… / Load every dial…** | The bottom bar's buttons. |

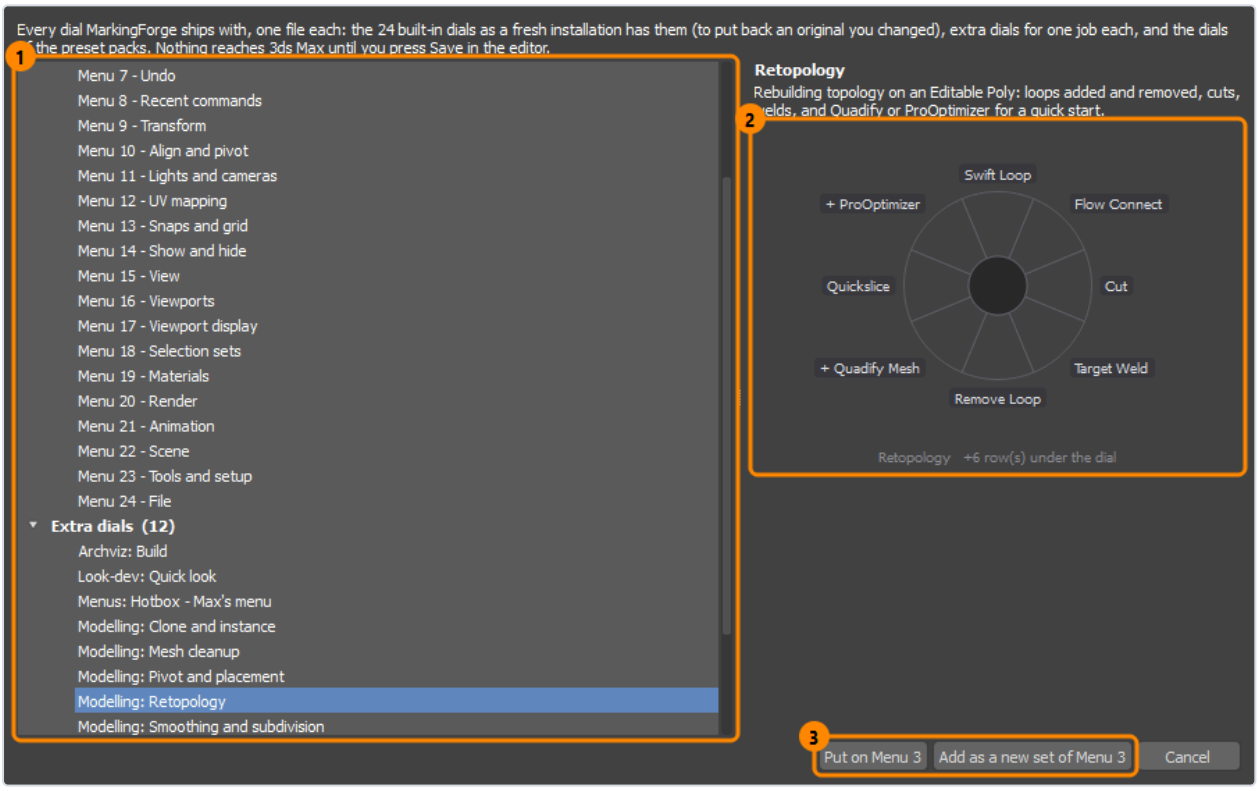
> **A file from somebody else can run code.** Scripts in a dial file, a folder of them or a preset run with full rights on your computer, so loading one removes them unless you say you trust the file. The dials of the Dial library ship with MarkingForge and keep theirs.

# 14. The quick ways: the library, the wizard and ready-made variants

### The dial library - every dial as a file of its own

MarkingForge ships every dial as a separate `.mfdial` file: the **24 built-in dials** exactly as a fresh installation has them, **12 extra dials** for one job each, and the **15 dials of the preset packs** one by one. **Dial library...** under the slot list - or *Put a dial from the library on Menu N...* in a dial's right-click menu - opens them, with a drawing of each.

1. Pick the dial you want to change in the slot list and press **Dial library...** - its built-in original is already selected.
2. **Restore Menu N** puts that original back - after you changed it and want it as it was.
3. Or pick any other dial on the left and press **Put on Menu N** (in place of what the dial shows; its key and mouse button stay) or **Add as a new set of Menu N** (the key keeps opening what it opens; the new set is its next page - the mouse wheel - and its tab opens here, chapter 12).
4. Save.




The dial library: built-in dials as shipped, extra dials and the packs' dials (1), the chosen dial (2), restore, put on the dial or add as a set (3).

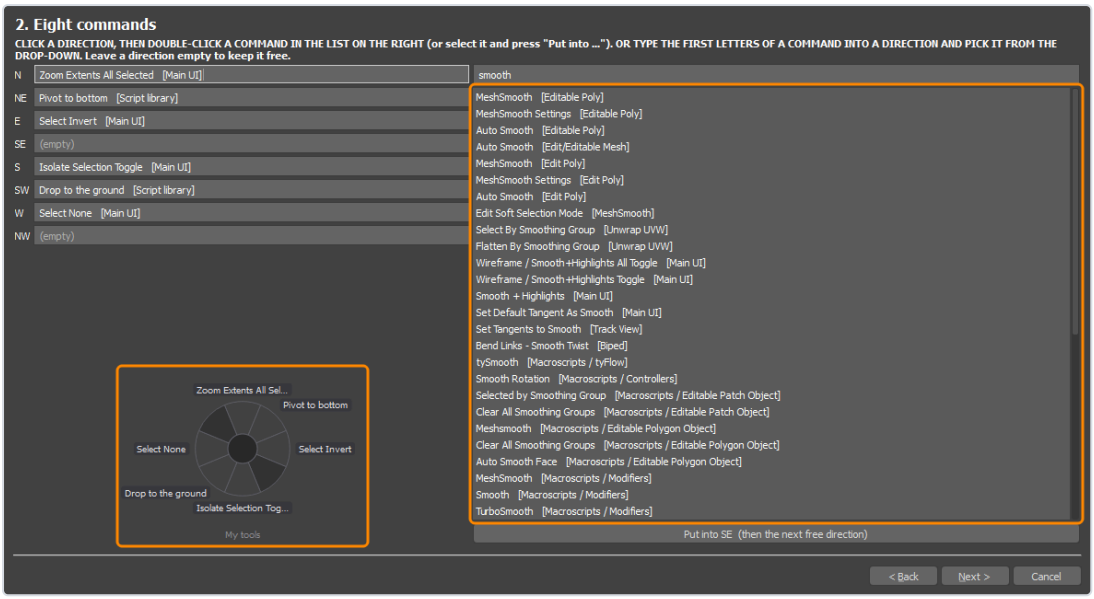
| Extra dial | What it is for |
|---|---|
| **Modelling - Mesh cleanup** | Find and fix what breaks smoothing and export: n-gons, triangles and holes on one side, the fixes on the other; reset the transform; reduce. |
| **Modelling - Pivot and placement** | Where an object's pivot is and where the object stands: pivot to the bottom or the centre, onto the ground, copy a transform from one object to another, align. |
| **Modelling - Clone and instance** | Every way to make more of an object: clone, array, spacing along a path, snapshot, mirror - and the instance tools on the other side. |

| | |
|---|---|
| **Modelling - Smoothing and subdivision** | Subdivision modifiers north and east, smoothing fixes south and west; switch every smoothing modifier of the selection off and on in one move. |
| **Modelling - Retopology** | Rebuilding topology on an Editable Poly: loops added and removed, cuts, welds, and Quadify or ProOptimizer for a quick start. |
| **Look-dev - Quick look** | Checking a model at a glance: grey clay, no material, random wire colours, see-through, box display and the UV checker. |
| **Rigging - Links and helpers** | Building a hierarchy: link and unlink, walk up and down it, freeze the transform, helpers to link to, and the deformers a rig starts with. |
| **Archviz - Build** | Architecture objects in 3ds Max's own creation modes: walls, doors, windows, stairs, railings and plants - drag them in the viewport as from the Create panel. |
| **Viewport - Viewport lighting** | Page 2 of Alt+5: shadows, ambient occlusion, highlights, scene or default lights, and materials with or without maps. |
| **Viewport - Viewport display** | Page 3 of Alt+5: grid, safe frames, statistics, ViewCube, selection display, isolate - and hiding whole categories in the list. |
| **Viewport - Viewport views** | Page 4 of Alt+5: top, front, left, right, perspective, orthographic, camera, maximize - zooms, layouts and the field of view in the list. |
| **Menus - Hotbox - Max's menu** | Max's whole main menu as tiles - the last page of Alt+5 (one wheel notch up; Alt+2 until 1.5.20), and until 1.4.4 the dial on its key itself. Put it on any key to have it there again. |

The same files are in the **Dials** folder of the download, for **Load dial...** as well.

### The new-dial wizard

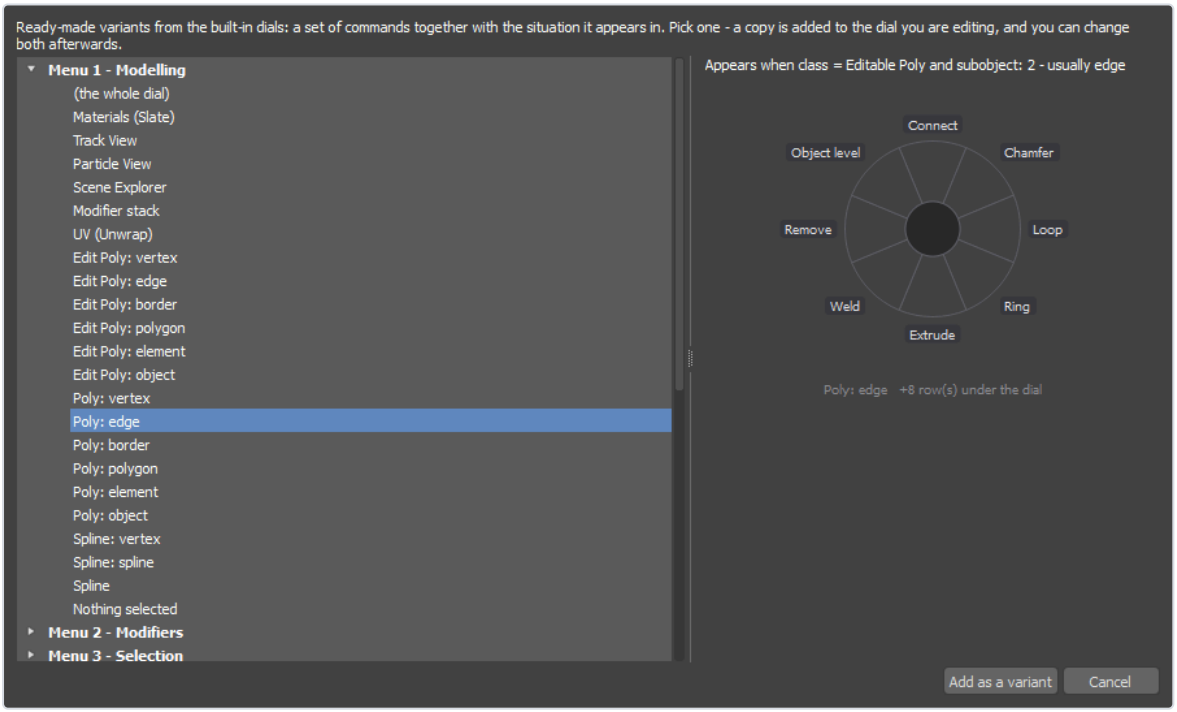
1. **New dial wizard...** under the slot list.
2. **Page 1** - which dial, its name, and whether the new contents become a **new set** (the default: the key keeps what it opens now, and the new set is the next page) or **replace** what the dial shows.
3. **Page 2** - eight fields, one per direction, and on the right a list of every 3ds Max command and the ready-made scripts with a search box. Click a direction, then double-click a command in the list (or select it and press **Put into ...**) - the next free direction is chosen for you. Or type the first letters of a command in a field and pick it from the drop-down; a name typed in full is found too (1.5.55). The drawing under the fields follows; a name that matches nothing turns red and is left out. A field left empty keeps its direction free.
4. **Page 3** - what will be made, with any name that was not found. Finish puts it in the editor; Save makes it real, and **Set shortcut...** gives the dial a key if it has none.




The wizard's second page.

### Ready-made variants - From a template...

Every context variant of the built-in dials is a tested set of commands with its condition: Editable Poly and Edit Poly level by level, splines, cameras, lights, the Slate, Track View and Scene Explorer windows. **From a template...** lists them all under their dials, with each built-in dial as a whole (*(the whole dial)*); the right side says when the chosen one appears and draws it. Pick one and press **Add as a variant**. It arrives with its condition (a whole dial asks you for one) at the end of the variant list - move it up if a more general variant above would catch it first.




The templates: what they are on the left, when they appear and what they hold on the right.

# 15. A good dial - a checklist

- **Eight is the limit for the hand.** What you rarely use goes to the list or a submenu; a different task gets a set of its own.
- **Opposites face each other** - hide / unhide, grow / shrink, in / out.
- **The same command in the same place** on every variant: Extrude north on every level teaches itself.
- **Captions of one or two words.** The dial is read at a glance.
- **Keep the caddy past the rim** for commands whose settings you change only sometimes.
- **A colour per dial** tells you which one opened before you read a word.
- **Watch the Usage column** after a week: a direction never used is a free place; one often cancelled is a command in the wrong place.
- **Save every dial** before big changes - Load every dial brings back every dial, its sets, the colours and every option, exactly as saved.

> **Still stuck?** Open an issue at github.com/Forge-Plugins/MarkingForge/issues or write to forgeplugins@gmail.com. **MarkingForge > Report a bug...** puts your 3ds Max and MarkingForge details on the clipboard, ready to paste.

# Support, bug reports and feature requests

| | |
|---|---|
| ISSUES | github.com/Forge-Plugins/MarkingForge/issues |
| ROADMAP | trello.com/b/7gJVqG8Y/markingforge-features-roadmap |
| E-MAIL | forgeplugins@gmail.com |
| WEB | https://forgeplugins.tech (forgeplugins.tech - more plugins by the author) |

**Bug reports and feature requests** are best opened as issues on GitHub - the forms there ask for exactly what is needed, and you can follow the fix. Questions, and anything you would rather not post publicly (licence, purchase), go to the e-mail address. Every message is read.

## Reporting a bug

1. In 3ds Max choose **MarkingForge > Report a bug...** - it copies your 3ds Max version, MarkingForge version, configuration summary and start-up log to the clipboard. It works even when the plugin did not load.
2. Open github.com/Forge-Plugins/MarkingForge/issues and pick **Bug report** - or write an e-mail to forgeplugins@gmail.com instead. Both are links in the window that opens (since 1.5.20), one click each.
3. Say what you did, what you expected and what happened instead, then paste the details with **Ctrl+V**.
4. Add a screenshot if you can. When the problem is in one dial, attach a preset of it (editor > **Save preset...**).

## Suggesting a feature

Open an issue with the **Feature request** form, choose **MarkingForge > Suggest a feature...**, or simply write. Describe what you would like to do and how you do it today - a concrete example from your work says more than a list of wishes. Ideas for new built-in dials, commands and options are all welcome. What is already planned and in progress is on the **Roadmap** above - **MarkingForge > Roadmap (Trello)** opens it too.

Nothing is sent by the plugin itself: the details go to the clipboard and your own e-mail program, so you see everything that leaves your computer. MarkingForge has no network access.